

UNIT-3: Network Layer

1. Network layer design issues.

2. Routing algorithms.

- a. Shortest path,
- b. Flooding,
- c. Distance vector,
- d. Link state routing,
- e. Hierarchical,
- f. Broadcast,
- g. Multicast, Anycast.

3. Congestion control algorithms.

4. Network layer in the internet –

The IP version 4 protocol, IP addresses, IP version 6,
Internet control protocols.

5. Open Shortest Path First (OSPF) & Border Gateway Protocol (BGP)

Introduction

- The **network layer** is concerned with getting packets from the source all the way to the destination.
- the network layer is the lowest layer that deals with **end-to-end** transmission.
- To achieve its goals,
 - the network layer must **know about the topology** of the communication subnet and choose appropriate paths through it.
 - It must also **take care to choose routes** to avoid overloading some of the communication lines and routers while leaving others idle.
 - Finally, when the **source and destination are in different** networks, new problems occur. It is up to the network layer to deal with them.

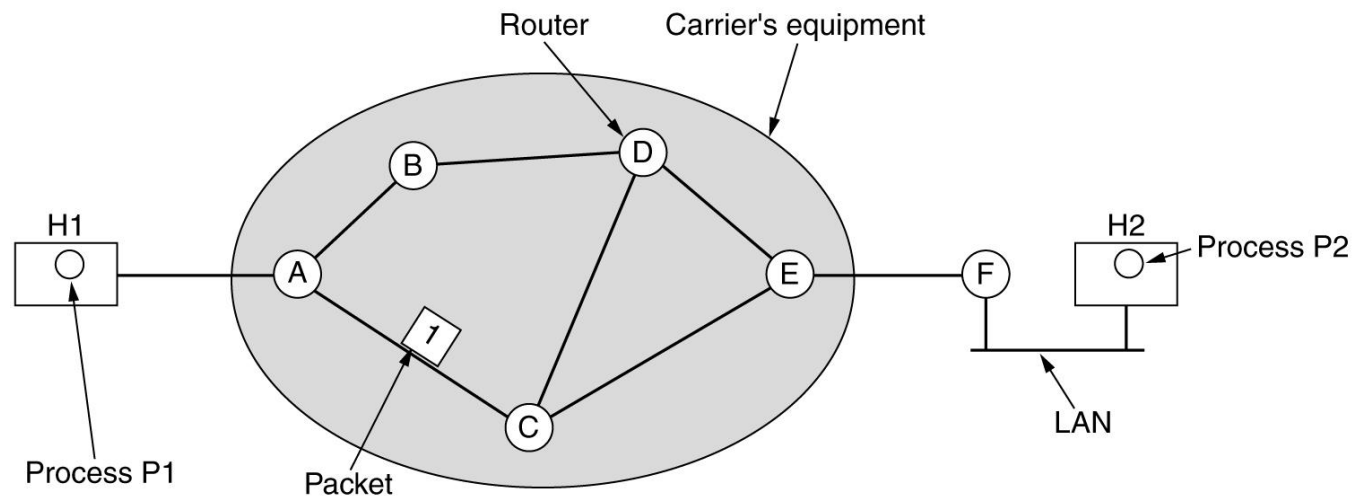
1. Network Layer Design Issues

These **issues include** the service provided to the transport layer and the internal design of the subnet.

1. Store-and-Forward Packet Switching
2. Services Provided to the Transport Layer
3. Implementation of Connectionless Service
4. Implementation of Connection-Oriented Service
5. Comparison of Virtual-Circuit and Datagram Subnets

1.1.Store-and-Forward Packet Switching

- But before starting to explain the details of the network layer, it is probably worth restating **the context in which the network layer protocols operate**.
- The major components of the system are the **carrier's equipment** (routers connected by transmission lines), shown inside the shaded oval, and the **customers' equipment**, shown outside the oval.



- This equipment is used as follows.
 - A host with a packet to send transmits it to the nearest router, either on its own LAN or over a point-to-point link to the carrier.
 - The packet is stored there until it has fully arrived so the checksum can be verified.
 - Then it is forwarded to the next router along the path until it reaches the destination host, where it is delivered.
 - This mechanism is **store-and-forward** packet switching.

1.2. Services Provided to the Transport Layer

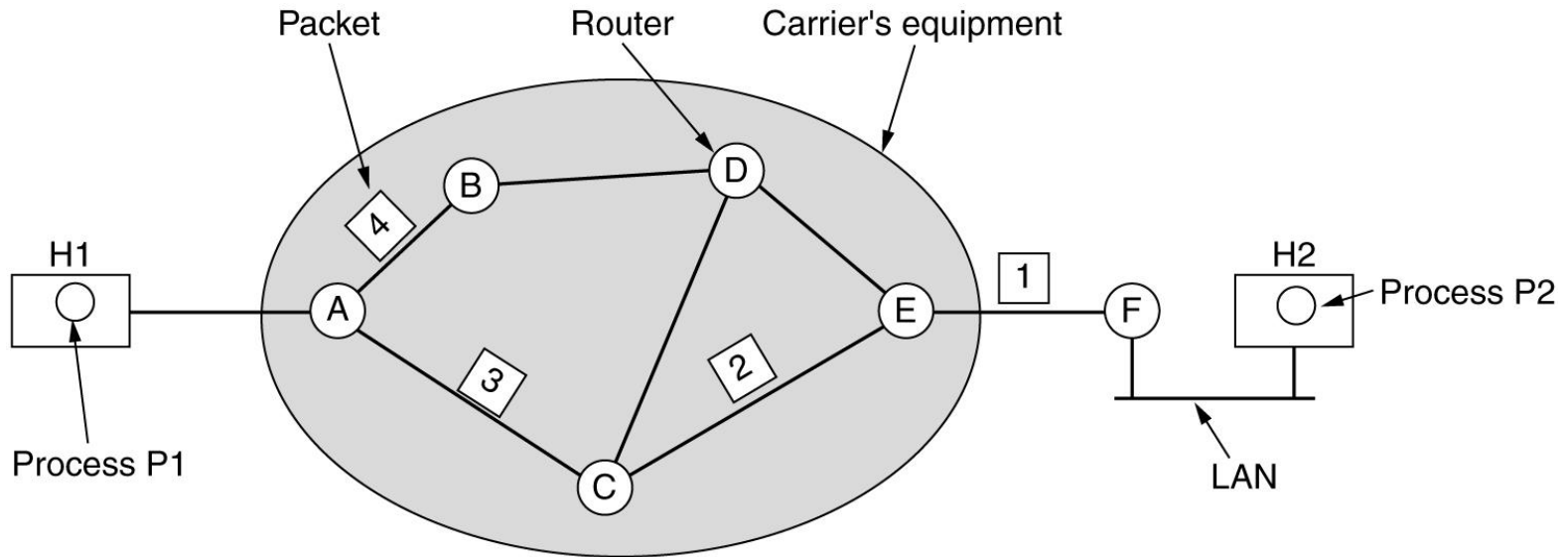
- The network layer provides services to the transport layer at the network layer/transport layer interface.
- An important question is what kind of services the network layer provides to the transport layer.
- The network layer services have been designed with the following goals in mind.
 1. The services should be independent of the router technology.
 2. The transport layer should be shielded from the number, type, and topology of the routers present.
 3. The network addresses made available to the transport layer should use a uniform numbering plan, even across LANs and WANs.

1. 3. Implementation of Connectionless Service

- Two different organizations are possible, depending on the type of service offered.
- If **connectionless service is offered**, packets are injected into the subnet individually and routed independently of each other.
- No advance setup is needed. In this context, the packets are frequently called datagrams (in analogy with telegrams) and the **subnet is called a datagram subnet**.
- If **connection-oriented service** is used, a path from the source router to the destination router must be established before any data packets can be sent.
- This **connection is called a VC (virtual circuit)**, in analogy with the physical circuits set up by the telephone system, and the subnet is called a **virtual-circuit subnet**.

Implementation of Connectionless Service

Routing within a diagram subnet.



A's table

initially

A	-
B	B
C	C
D	B
E	C
F	C

later

A	-
B	B
C	C
D	B
E	B
F	B

C's table

A	A
B	A
C	-
D	D
E	E
F	E

E's table

A	C
B	D
C	C
D	D
E	-
F	F

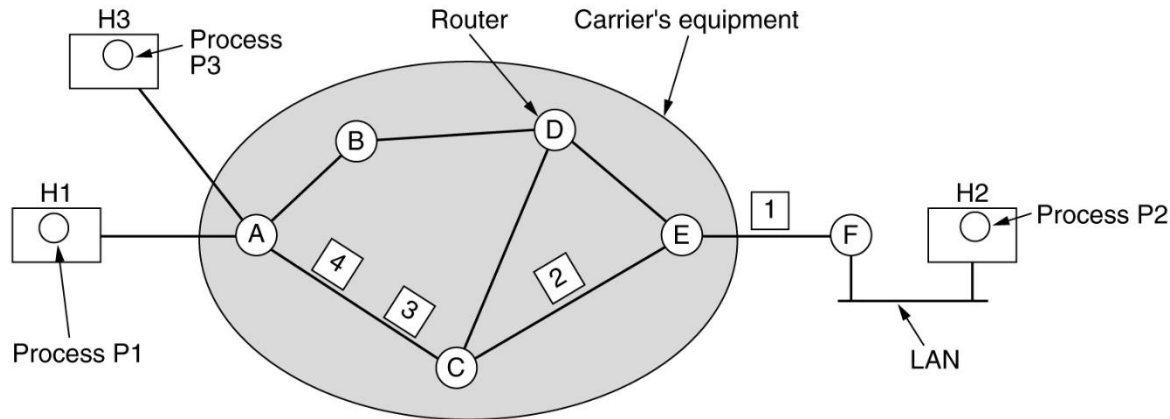
Dest. Line

1.4. Implementation of Connection-Oriented Service

- For connection-oriented service, we need a virtual-circuit subnet.
- The idea behind virtual circuits is to **avoid having to choose a new route** for every packet sent.
- when a connection is established, a route from the source machine to the destination machine is chosen as part of the connection setup and **stored in tables inside the routers**.
- That route is used for all traffic flowing over the connection, exactly the same way that the **telephone system works**.
- When the connection is released, the virtual circuit is also terminated. With connection-oriented service, each packet carries an identifier telling which virtual circuit it belongs to.

Implementation of Connection-Oriented Service

Routing within a virtual-circuit subnet.



A's table				C's table				E's table			
H1	1	C	1	A	1	E	1	C	1	F	1
H3	1	C	2	A	2	E	2	C	2	F	2
In		Out									

- Here, host H1 has established connection 1 with host H2.
- It is remembered as the first entry in each of the routing tables.
- The first line of A's table says that if a packet bearing connection identifier 1 comes in from H1, it is to be sent to router C and given **connection identifier 1**.
- Similarly, the first entry at C routes the packet to E, also with connection identifier 1.

- Now let us consider what happens if H3 also wants to establish a connection to H2.
- It chooses connection identifier 1 (because it is initiating the connection and this is its only connection) and tells the subnet to establish the virtual circuit.
- This leads to the second row in the tables. Note that we have a conflict here because although A can easily distinguish connection 1 packets from H1 from connection 1 packets from H3, C cannot do this.
- For this reason, A assigns a different connection identifier to the outgoing traffic for the second connection.
- Avoiding conflicts of this kind is why routers need the ability to replace connection identifiers in outgoing packets. In some contexts, this is called **label switching**.
- **An example of a connection oriented network service is MPLS (Multi Protocol Label Switching).**

1. 5. Comparison of Virtual-Circuit and Datagram Subnets

Issue	Datagram subnet	Virtual-circuit subnet
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

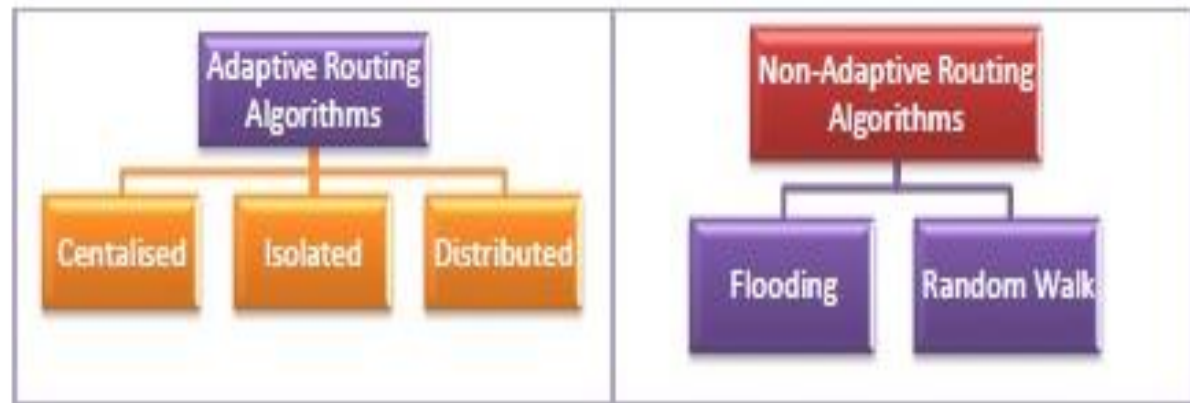
2. Routing Algorithms:

- 1. Shortest path**
- 2. Flooding, Distance vector**
- 3. Link state routing**
- 4. Hierarchical**
- 5. Broadcast**
- 6. Multicast**
- 7. Anycast.**

2. Routing Algorithms

- ✓ The routing algorithm is that part of the network layer software responsible for deciding which output line an incoming packet should be transmitted on.
- ✓ The latter case is sometimes called session routing because a route remains in force for an entire session (e.g., while logged in over a VPN).
- ✓ **Routing algorithms can be grouped into two major classes:**
Non adaptive and Adaptive.
- ✓ **Non adaptive algorithms** do not base their routing decisions on any measurements or estimates of the current topology and traffic.
- ✓ Instead, the choice of the route to use to get from I to J (for all I and J) is computed in advance, offline, and downloaded to the routers when the network is booted, this procedure is sometimes called static routing. It consists of

- a) The Optimality Principle
- b) Shortest Path Routing
- c) Flooding
- d) Distance Vector Routing
- e) Link State Routing
- f) Hierarchical Routing
- g) Broadcast Routing
- h) Multicast Routing



- **Adaptive routing algorithm** is also called as **dynamic routing algorithm**.
- In this algorithm, the routing decisions are made based on network traffic and topology.
- The parameters which are used in adaptive routing algorithms are distance, hop, estimated transit time and count.
- **Non-adaptive routing algorithm** is also called as **static routing algorithm**.
- In a non-adaptive routing algorithm, the routing decisions are not made based on network traffic and topology.
- This algorithm is used by static routing. Non-adaptive routing algorithms are simple as compared to Adaptive routing algorithm in terms of complexity.

Difference between Adaptive and Non-Adaptive Routing algorithms

S.NO	Adaptive Routing algorithm	Non-Adaptive Routing algorithm
1.	This algorithm creates a routing table based on network conditions.	Whereas this algorithm creates a static table in order to determine when to send packets and which node.
2.	This algorithm is used by dynamic routing.	Whereas this algorithm is used by static routing.
3.	In adaptive routing algorithm, the routing decisions are made based on network traffic and topology.	Whereas in a non-adaptive routing algorithm, the routing decisions are not made based on network traffic and topology.
4.	Adaptive routing algorithms are more complex as compared to non-adaptive routing algorithms in terms of complexity.	While non-adaptive routing algorithms are simple in terms of complexity.
5.	In adaptive routing algorithm, the routing decisions are not static tables.	While in non-adaptive routing algorithm, the routing decisions are static tables.
6.	Adaptive routing algorithm is categorized into distributed, centralized and isolation algorithm.	Whereas non-adaptive routing algorithm is categorized into random walks and flooding.
7.	Adaptive routing algorithm is more used as compared to non-adaptive.	Whereas non-adaptive routing algorithm is comparatively less used.

- ✓ **Adaptive algorithms**, in contrast, change their routing decisions to reflect changes in the topology, and sometimes changes in the traffic as well. These dynamic routing algorithms differ in where

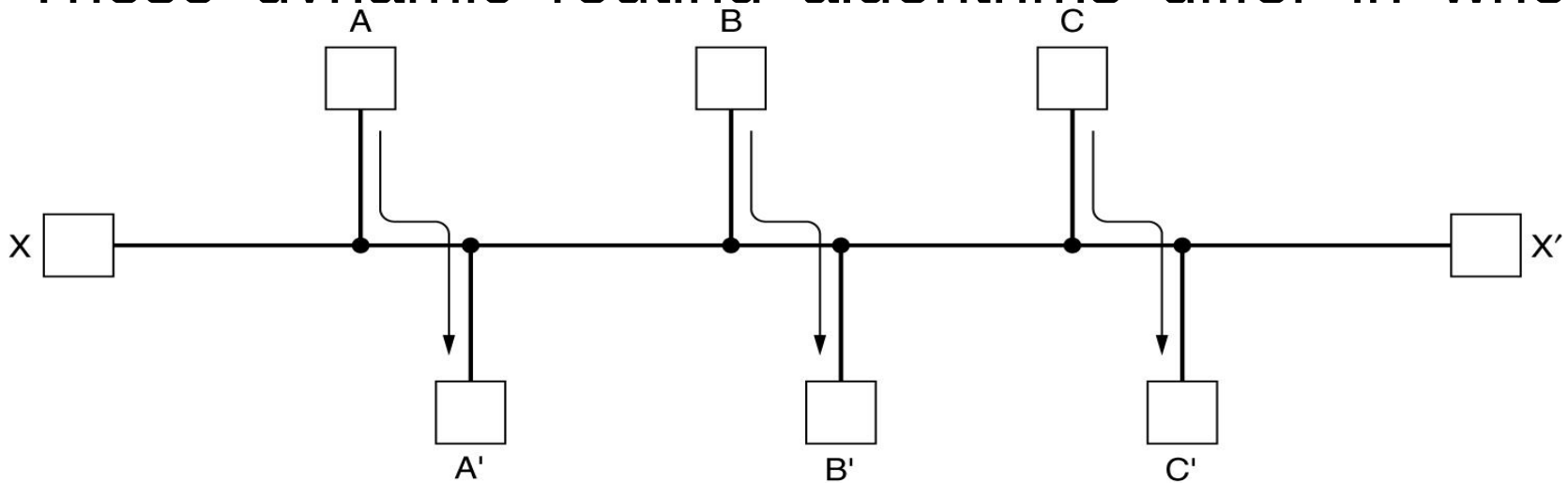


Fig 5-5: Network with a conflict between fairness and efficiency.

- ✓ For example, router F in Fig. 5-3 should send packets headed into the network to router E regardless of the ultimate destination.

a) The Optimality Principle: Specific algorithms, it may be helpful to note that one can make a general statement about optimal routes without regard to network topology or traffic. This statement is known as the **optimality principle**.

- ✓ The set of optimal routes from all sources to a given destination form a tree rooted at the destination. Such a tree is called a sink tree and is illustrated in Fig. 5-6(b).
- ✓ The goal of all routing algorithms is to discover and use the sink trees for all routers.
- ✓ Note that a sink tree is not necessarily unique; other trees with the same path lengths may exist.
- ✓ If we allow all of the possible paths to be chosen, the tree becomes a more general structure called a **DAG (Directed Acyclic Graph)**. DAGs have no loops.

- ✓ Links and routers can go down and come back up during operation, so different routers may have different ideas about the current topology.

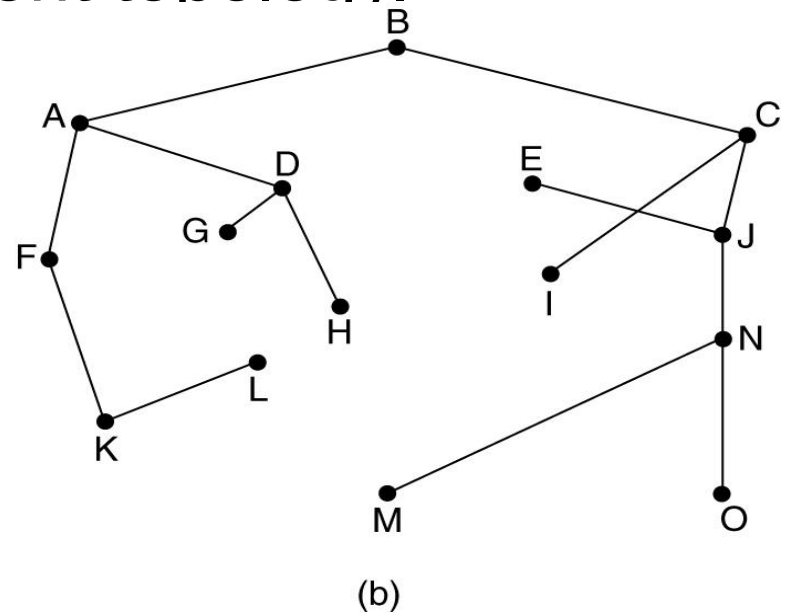
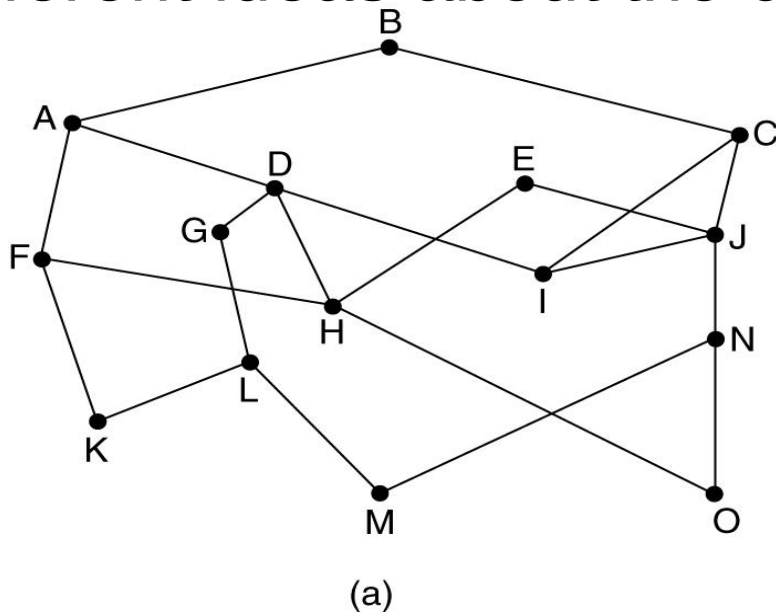


Fig 5-6. (a) A network. (b) A sink tree for router B.

b) Shortest Path Algorithm: The concept of a shortest path deserves some explanation. One way of **measuring path length is the number of hops**. Using this metric, the paths ABC and ABE in Fig. 5-7 are equally long.

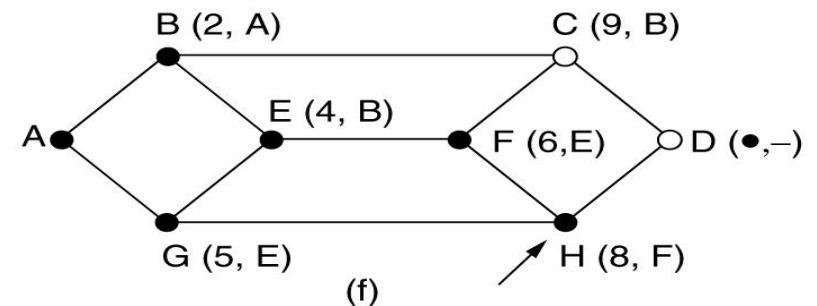
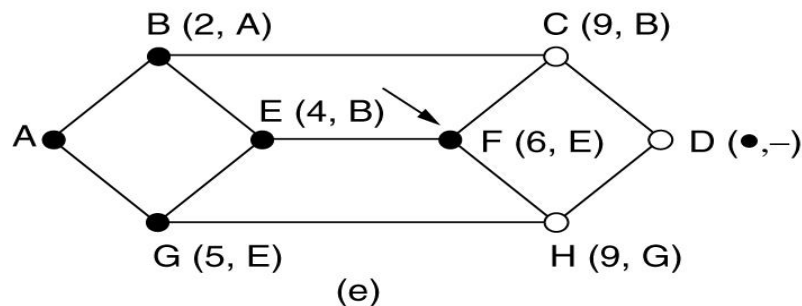
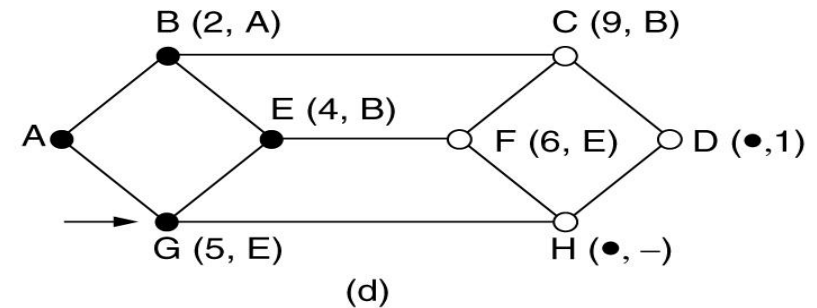
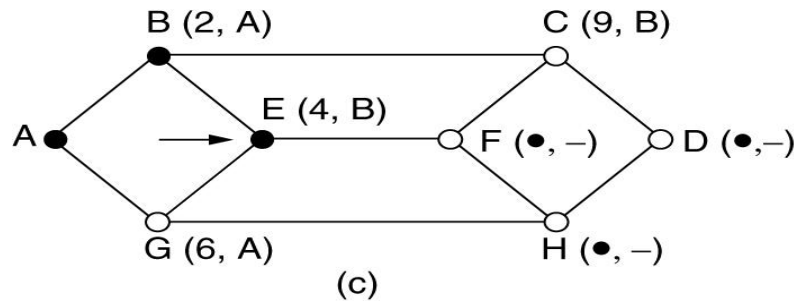
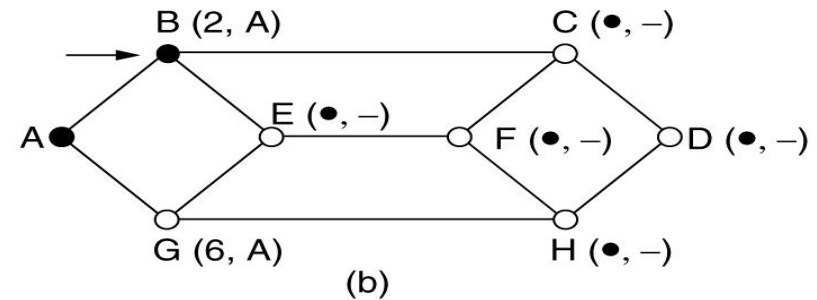
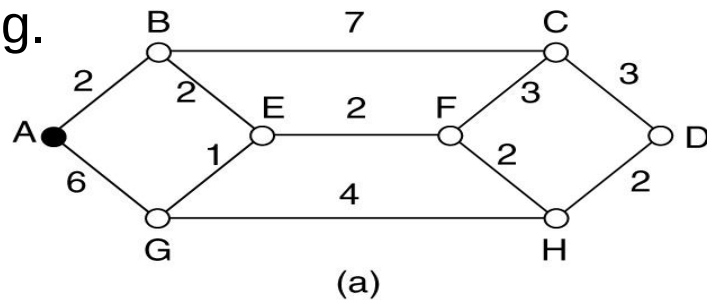


Fig.5-7. The first six steps used in computing the shortest path from A to D. The arrows indicate the working node.

- ✓ Several algorithms for computing the shortest path between two nodes of a graph are known. This one is due to **Dijkstra (1959)** and finds the shortest paths between a source and all destinations in the network.
- ✓ To illustrate how the labeling algorithm works, look at the **weighted, undirected graph** of Fig. 5-7(a), where the **weights represent distance**.
- ✓ We want to find the shortest path from A to D. Having examined each of the nodes adjacent to A, we examine all the tentatively labeled nodes in the whole graph and make the one with the smallest label permanent, as shown in Fig. 5-7(b).
- ✓ To see why the algorithm works, look at Fig. 5-7(c). At this point we have just made E permanent.
- ✓ There are two possibilities: either node Z has already been made permanent, or it has not been. If it has, then E has already been probed so the AXYZE path has not escaped our attention and thus cannot be a shorter path.

c) Flooding: A simple local technique is flooding, in which every **incoming packet is sent out on every outgoing line except the one it arrived on.**

- ✓ Flooding obviously generates **vast numbers of duplicate packets**, in fact, an infinite number unless some measures are taken to damp the process.
- ✓ Flooding with a **hop count** can produce an exponential number of duplicate packets as the hop count grows and routers duplicate packets they have seen before.
- ✓ A better technique for damming the flood is to have routers keep track of which packets have been flooded, to avoid sending them out a second time.

d) Distance Vector Routing: Distance vector routing algorithm operates by having each router maintain a table (i.e., a vector) giving the best known distance to each destination and which link to use to get there.

✓ The distance vector routing algorithm is sometimes called by other names, most commonly the distributed **Bellman-Ford routing algorithm**.

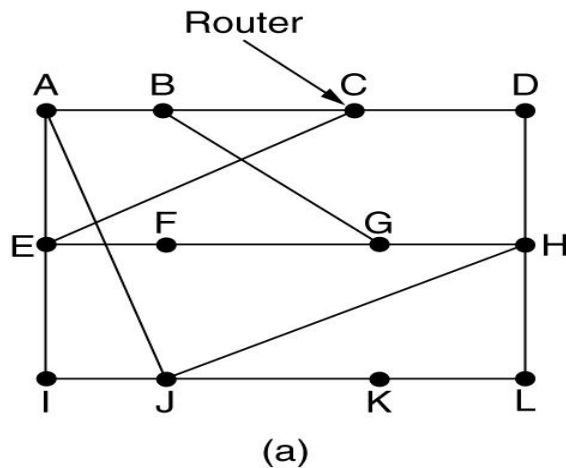


Fig 5-9: (a) A subnet. **(b)** Input from A, I, H, K, and the new routing table for J.

To	A	I	H	K	New estimated delay from J ↓ Line	
A	0	24	20	21	8	A
B	12	36	31	28	20	A
C	25	18	19	36	28	I
D	40	27	8	24	20	H
E	14	7	30	22	17	I
F	23	20	19	40	30	I
G	18	31	6	31	18	H
H	17	20	0	19	12	H
I	21	0	14	22	10	I
J	9	11	7	10	0	—
K	24	22	22	0	6	K
L	29	33	9	9	15	K

JA delay is 8 JI delay is 10 JH delay is 12 JK delay is 6

Vectors received from J's four neighbors

New routing table for J

(b)

- ✓ **The Count-to-Infinity Problem:** The settling of routes to best paths across the network is called **convergence** (coming together).
- ✓ Distance vector routing is useful as a simple technique by which routers can collectively compute shortest paths.
- ✓ To see how fast good news propagates, consider the five-node (linear) network of Fig. 5-10, where the delay metric is the number of hops.

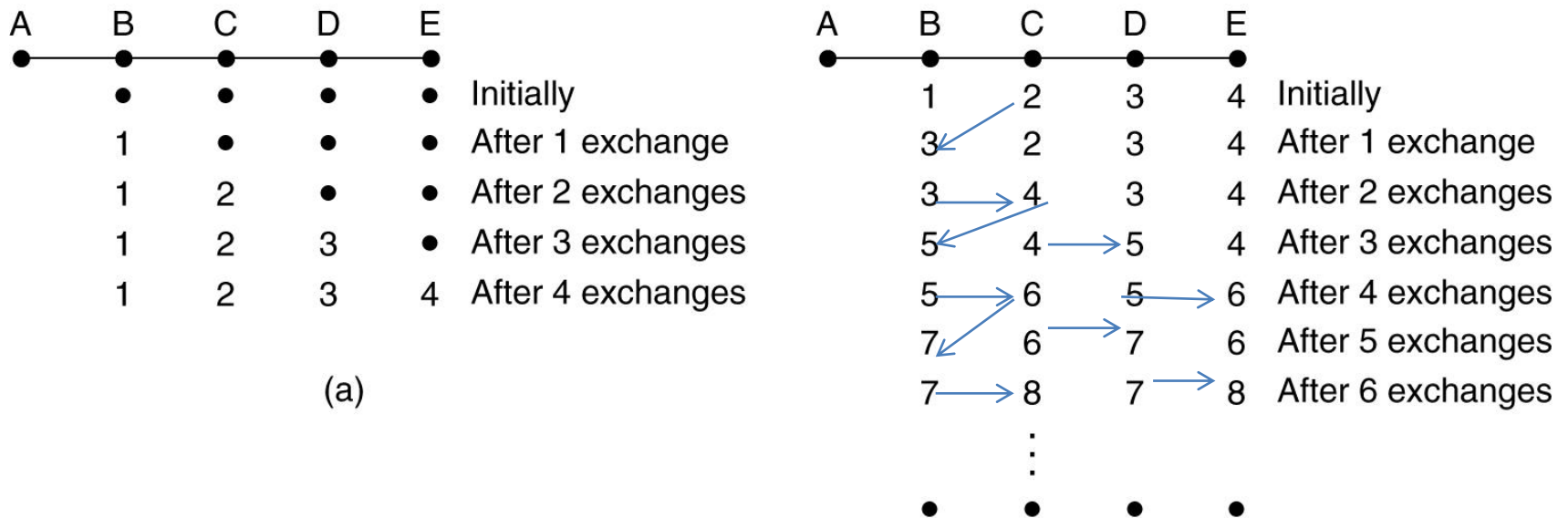


Figure 5-10. The count-to-infinity problem (Routing Loop).

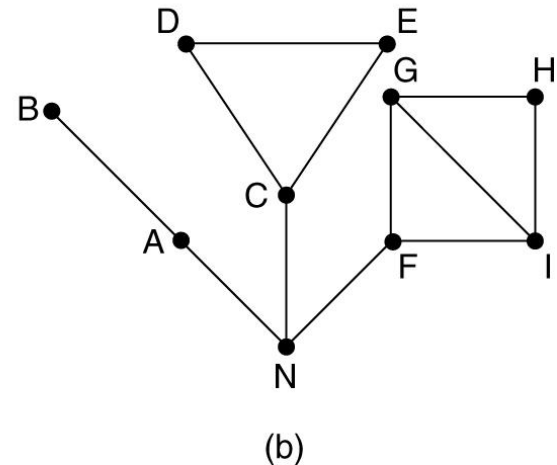
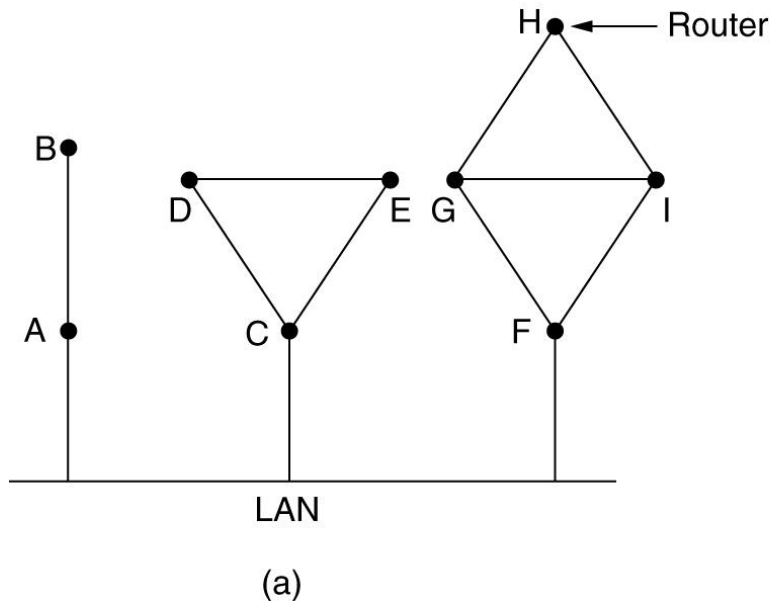
- ✓ Each router must do the following:
 1. Discover its neighbors, learn their network address.
 2. Measure the delay or cost to each of its neighbors.
 3. Construct a packet telling all it has just learned.
 4. Send this packet to all other routers.
 5. Compute the shortest path to every other router.

e) **Link State Routing:** The primary problem that caused its demise was that the algorithm often took too long to converge after the network topology changed (due to the count-to-infinity problem).

- ✓ Consequently, it was replaced by an entirely new algorithm, now called **Link State Routing**.
- ✓ The idea behind link state routing is fairly simple and can be stated as **five parts**.
- ✓ Each router must do the following things to make it work:
 1. Discover its neighbors and learn their network addresses.
 2. Set the distance or cost metric to each of its neighbors.
 3. Construct a packet telling all it has just learned.
 4. Send this packet to and receive packets from all other routers.
 5. Compute the shortest path to every other router.

Learning about the Neighbors

- ✓ When two or more routers are connected by a broadcast link (e.g., a switch, ring, or classic Ethernet), the situation is slightly more complicated. Fig. 5-11(a) illustrates a broadcast LAN to which three routers, A, C, and F, are directly connected.
- ✓ A better way to model the LAN is to consider it as a node itself, as shown in Fig. 5-11(b).
- ✓ Fig: 5.11(a) Nine routers and a LAN. (b) A graph model of (a).



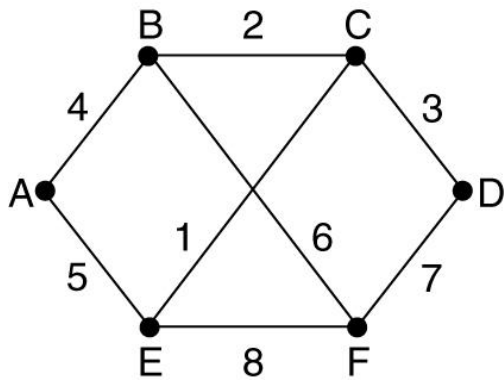
Setting Link Costs

- ✓ The link state routing algorithm requires each link to have a distance or cost metric for finding shortest paths.
- ✓ The most direct way to determine this delay is to send over the line a special ECHO packet that the other side is required to send back immediately.
- ✓ By measuring the round-trip time and dividing it by two, the sending router can get a reasonable estimate of the delay.

Building Link State Packets

- ✓ The packet starts with the identity of the sender, followed by a sequence number and age and a list of neighbors.
- ✓ The cost to each neighbor is also given. An example network is presented in Fig. 5-12(a) with costs shown as labels on the lines. The corresponding link state packets for all six routers are shown in Fig. 5- 12(b).
- ✓ Building the link state packets is easy.

- ✓ The hard part is determining when to build them. One possibility is to build them periodically, that is, at regular intervals.
- ✓ Another possibility is to build them when some significant event occurs, such as a line or neighbor going down or coming back up again or changing its properties appreciably.



(a)

A		B		C		D		E		F	
Seq.		Seq.		Seq.		Seq.		Seq.		Seq.	
Age		Age		Age		Age		Age		Age	
B	4	A	4	B	2	C	3	A	5	B	6
E	5	C	2	D	3	F	7	C	1	D	7
		F	6	E	1			F	8	E	8

(b)

Fig:5-12. (a) A network.

(b) The link state packets for this network.

Distributing the Link State Packets

- ✓ The trickiest part of the algorithm is distributing the link state packets.
- ✓ If different routers are using different versions of the topology, the routes they compute can have inconsistencies such as loops, unreachable machines, and other problems.
- ✓ First, the basic distribution algorithm. After that we will give some refinements. The fundamental idea is to use flooding to distribute the link state packets to all routers.
- ✓ Second, if a router ever crashes, it will lose track of its sequence number. If it starts again at 0, the next packet it sends will be rejected as a duplicate.
- ✓ Third, if a sequence number is ever corrupted and 65,540 is received instead of 4 (a 1-bit error), packets 5 through 65,540 will be rejected as obsolete, since the current sequence number will be thought to be 65,540

- ✓ The data structure used by router B for the network shown in Fig. 5-12(a) is depicted in Fig. 5-13.

Source	Seq.	Age	Send flags			ACK flags			Data
			A	C	F	A	C	F	
A	21	60	0	1	1	1	0	0	
F	21	60	1	1	0	0	0	1	
E	21	59	0	1	0	1	0	1	
C	20	60	1	0	1	0	1	0	
D	21	59	1	0	0	0	1	1	

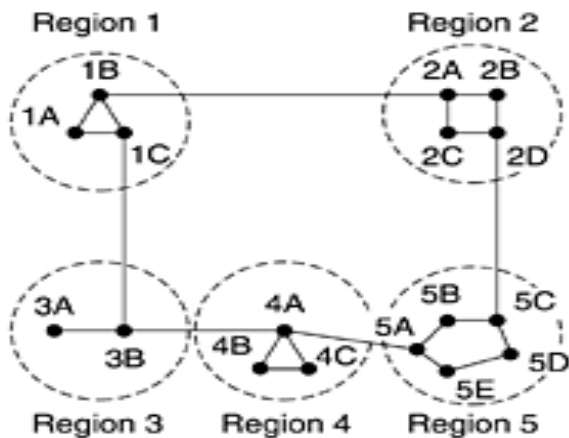
Fig 5-13: The packet buffer for router B in the previous slide

Computing the New Routes

- ✓ Link state routing is widely used in actual networks, so a few words about some example protocols are in order. Many ISPs use the IS-IS (Intermediate System-Intermediate System) link state protocol (Oran, 1990). It was designed for an early network called DECnet.

Hierarchical Routing

- As networks grow in size, the router routing tables grow proportionally.
- Not only is router memory consumed by ever-increasing tables, but more CPU time is needed to scan them and more bandwidth is needed to send status reports about them.
- At a certain point the network may grow to the point where it is no longer feasible for every router to have an entry for every other router, so the routing will have to be done hierarchically.
- When hierarchical routing is used, the routers are divided into what we will call regions, with each router knowing all the details about how to route packets to destinations within its own region, but knowing nothing about the internal structure of other regions.
- For huge networks, a two-level hierarchy may be insufficient; it may be necessary to group the regions into clusters, the clusters into zones, the zones into groups, and so on, until we run out of names for aggregations.



(a)

Full table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

(b)

Hierarchical table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

(c)

gives a quantitative example of routing in a two-level hierarchy with five regions.

The full routing table for router *1A* has 17 entries, as shown in Fig. (b). Hierarchical routing has reduced the table from 17 to 7 entries.

As the ratio of the number of regions to the number of routers per region grows, the savings in table space increase.

5. Broadcast Routing

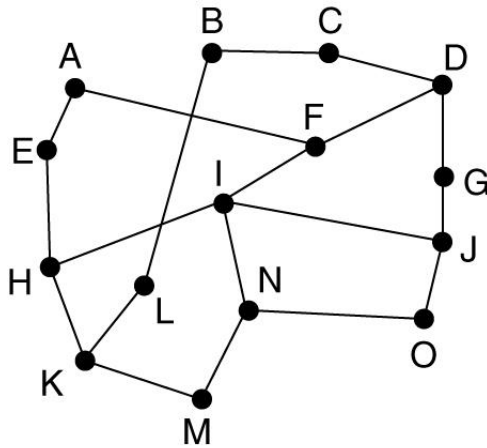
- In some applications, hosts need to send messages to many or all other hosts.
- For example, a service distributing **weather reports, stock market updates**, or **live radio** programs might work best by broadcasting to all machines and letting those that are interested read the data.
- Sending a packet to all destinations simultaneously is called **broadcasting**.
- **various methods** have been proposed for doing it.
- **Methods:**
 - **simply send a distinct packet to each destination.**
 - Not only is the method wasteful of bandwidth, but it also requires the source to have a complete list of all destinations.
 - **Flooding**
 - ill-suited for ordinary point-to-point communication.
 - it generates too many packets and consumes too much bandwidth.
 - A third algorithm is **multi destination** routing.
 - Each packet contains either a list of destinations or a bit map indicating the desired destinations.
 - When a packet arrives at a router, the router checks all the destinations to determine the set of output lines that will be needed.

- **Sink tree**

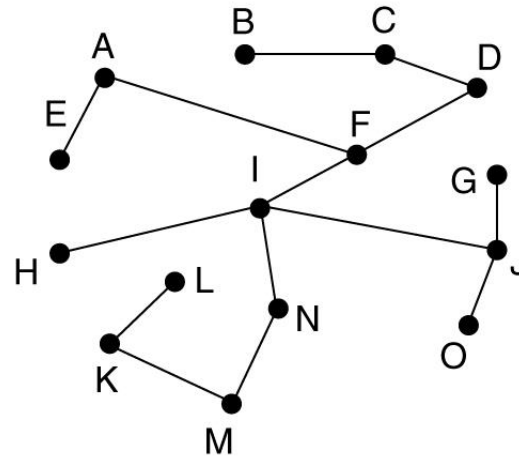
- A spanning tree is a subset of the subnet that includes all the routers but contains no loops.
- If each router knows which of its lines belong to the spanning tree, it can copy an incoming broadcast packet onto all the spanning tree lines except the one it arrived on.
- This method makes **excellent use of bandwidth**, generating the absolute minimum number of packets necessary to do the job.
- The only problem is that **each router must have knowledge** of some spanning tree for the method to be applicable.

- Our last broadcast algorithm is **an attempt to approximate the behavior of the previous one**, even when the routers do not know anything at all about spanning trees.
- The idea, called **reverse path forwarding**.
- When a broadcast packet arrives at a router, the router checks to see if the packet arrived on the line that is normally used for sending packets to the source of the broadcast.
- If so, there is an excellent chance that the broadcast packet itself followed the best route from the router and is therefore the first copy to arrive at the router.
- This being the case, the router forwards copies of it onto all lines except the one it arrived on.
- If, however, the broadcast packet arrived on a line other than the preferred one for reaching the source, the packet is discarded as a likely duplicate.

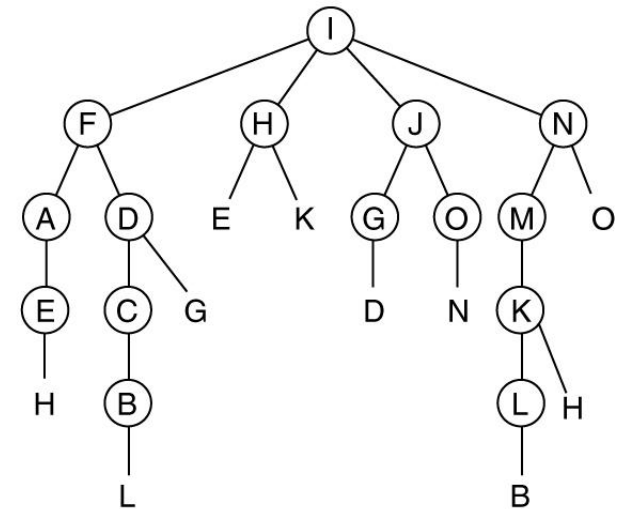
Broadcast Routing



(a)



(b)



(c)

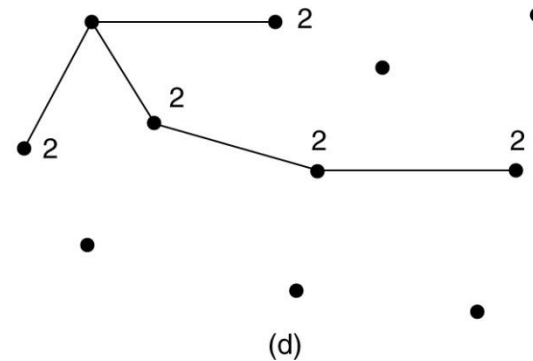
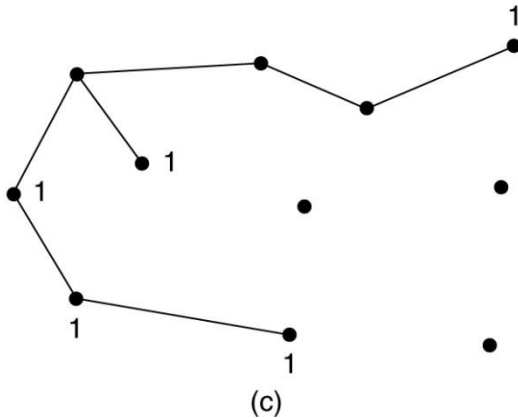
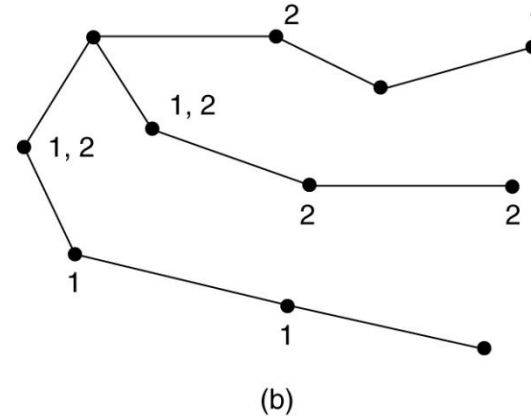
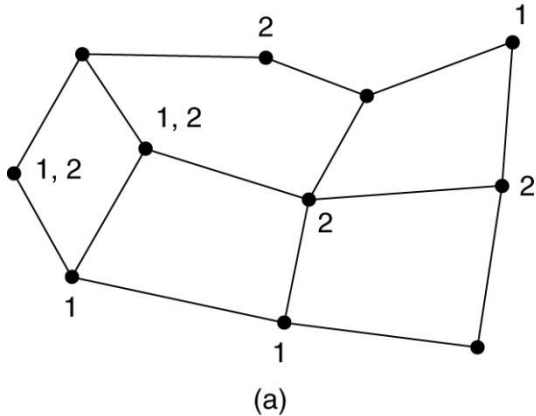
Reverse path forwarding. (a) A subnet. (b) a Sink tree. (c) The tree built by reverse path forwarding.

- The principal **advantage** of reverse path forwarding is that it is both reasonably efficient and easy to implement.
- It does **not require routers to know about spanning trees**, nor does it have the overhead of a destination list or bit map in each broadcast packet as does multdestination addressing.
- **Nor does it require any special mechanism** to stop the process, as flooding does

6. Multicast Routing

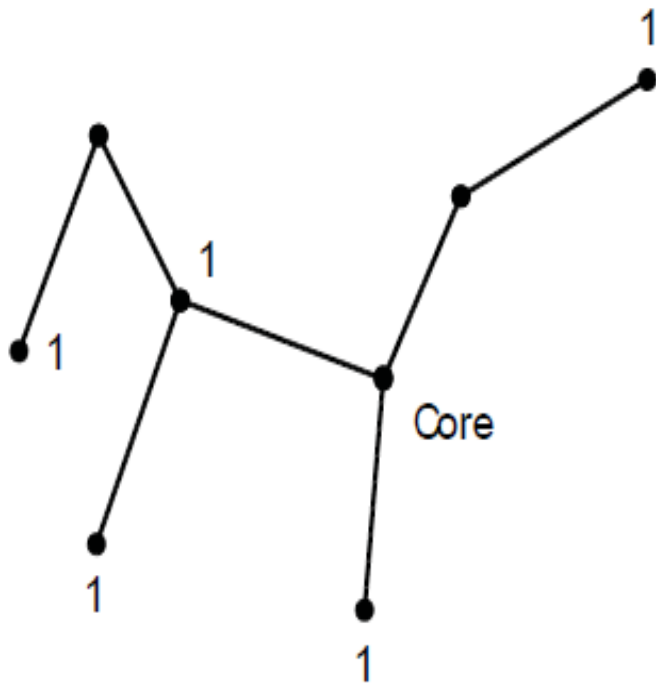
- Sending a message to such a group is called **multicasting**, and its routing algorithm is called **multicast routing**.
- Multicasting requires **group management**. Some way is needed to create and destroy groups, and to allow processes to join and leave groups.
- How **these tasks are accomplished is not of concern** to the routing algorithm.
- What is of concern is that when a process joins a group, it informs its host of this fact.
- It is **important that routers know which of their hosts belong to which groups**. Either **hosts must inform their routers** about changes in group membership, or **routers must query their hosts periodically**. Either way, routers learn about which of their hosts are in which groups. Routers tell their neighbors, so the information propagates through the subnet.
- To do multicast routing, **each router computes a spanning tree covering** all other routers.
- When a process sends a multicast packet to a group, the first router examines its spanning tree and prunes it, removing all lines that do not lead to hosts that are members of the group.
- **Various ways of pruning the spanning tree are possible.**
 - link state routing
 - distance vector routing

Multicast Routing

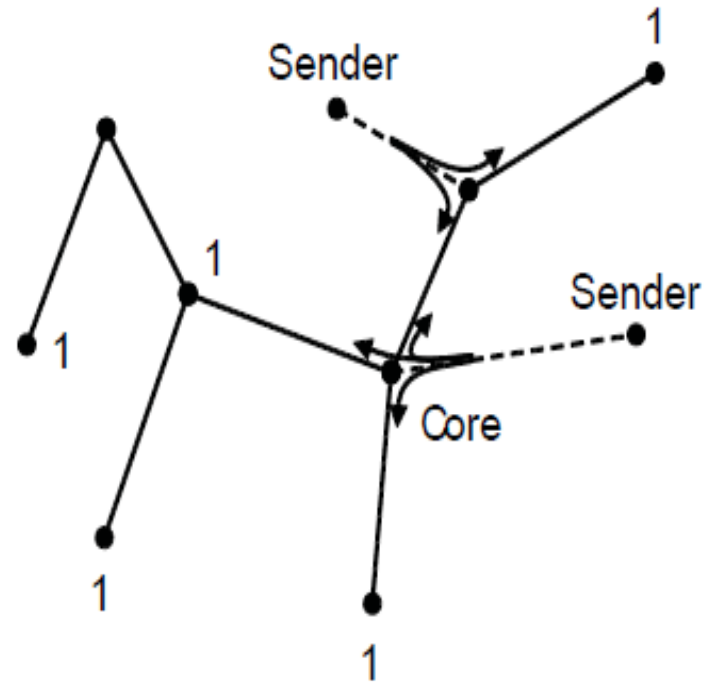


- (a) A network. (b) A spanning tree for the leftmost router.
 (c) A multicast tree for group 1. (d) A multicast tree for group 2.

- One potential **disadvantage** of this algorithm is that it **scales poorly** to large networks.
- Suppose that a network has **n** groups, each with an average of **m** members.
- For each group, **m** pruned spanning trees must be stored, for a total of **mn** trees.
- When many large groups exist, considerable storage is needed to store all the trees.
- An alternative design uses **core-based trees**
- Here, a single spanning tree per group is computed, with the root (the **core**) near the middle of the group.
- To send a multicast message, a host sends it to the core, which then does the multicast along the spanning tree.
- Although this tree will not be optimal for all sources, the reduction in storage costs from **m** trees to one tree per group is a major saving.



(a)



(b)

(a) Core-based tree for group 1.

(b) Sending to group 1.

Anycast Routing: we have covered delivery models in which a source sends to a single destination (called unicast), to all destinations (called broadcast), and to a group of destinations (called multicast).

- ✓ Another delivery model, called anycast is sometimes also useful. In anycast, a packet is delivered to the nearest member of a group (Partridge et al., 1993). Schemes that find these paths are called anycast routing.
- ✓ The routes are shown in Fig. 5-18(a). This procedure works because the routing protocol does not realize that there are multiple instances of destination 1.
- ✓ That is, it believes that all the instances of node 1 are the same node, as in the topology shown in Fig. 5-18(b).

- ✓ This procedure works for link state routing as well, although there is the added consideration that the routing protocol must not find seemingly short paths that pass through node 1.

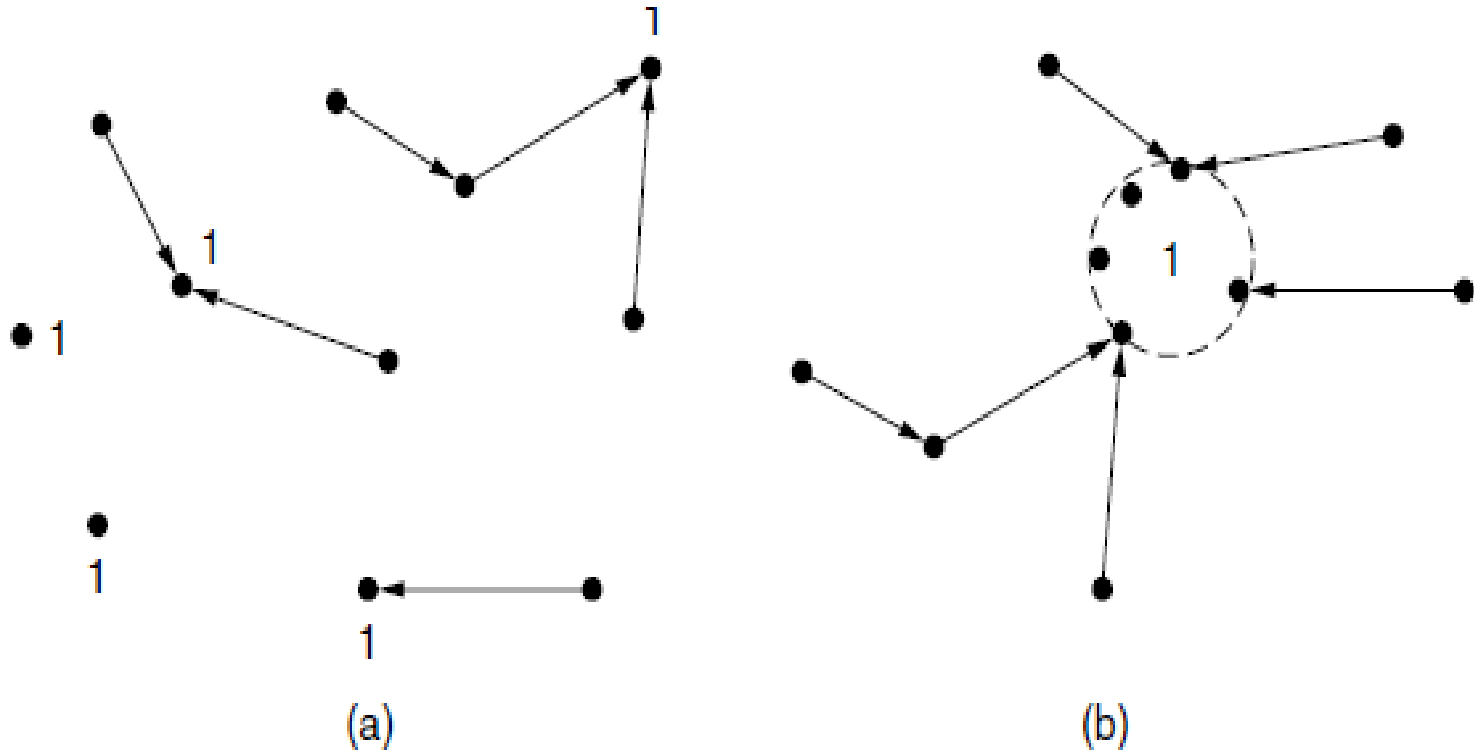


Fig.5-18. (a) Anycast routes to group 1.
(b) Topology seen by the routing protocol.

III. Congestion Control Algorithms

- When too many packets are present in (a part of) the subnet, performance degrades. This situation is called **congestion**.

Congestion Prevention Policies



CONGESTION CONTROL ALGORITHMS: Too many packets present in (a part of) the network causes packet delay and loss that degrades performance.

- ✓ This situation is called **congestion**. This requires the network and transport layers to work together.
- ✓ Figure 5-21 depicts the onset of congestion. When the number of packets hosts send into the network is well within its carrying capacity, the number delivered is proportional to the number sent.

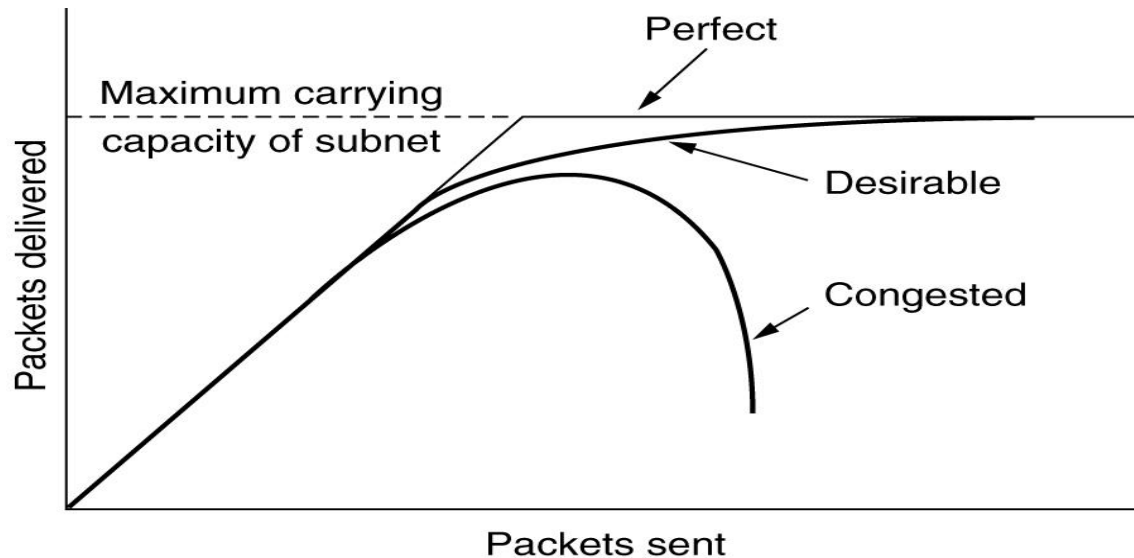


Figure 5-21. With too much traffic, performance drops sharply.

Congestion Control Algorithms: It consists of

- General Principles of Congestion Control
- Congestion Prevention Policies
- Congestion Control in Virtual-Circuit Subnets
- Congestion Control in Datagram Subnets
- Load Shedding
- Jitter Control

1. Approaches to Congestion Control

- ✓ The presence of congestion means that the load is (temporarily) greater than the resources (in a part of the network) can handle. **Two solutions come to mind: increase the resources or decrease the load.**
- ✓ As shown in Fig. 5-22, these solutions are usually applied on different time scales to either prevent congestion or react to it once it has occurred.

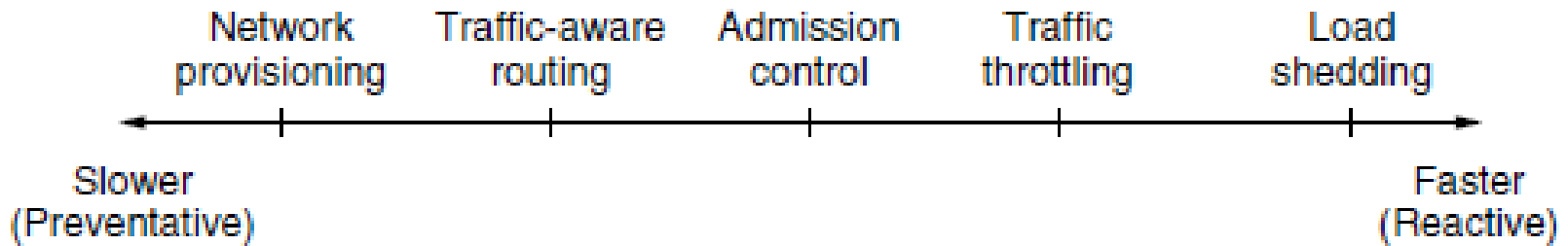


Fig.5-22. Timescales of approaches to congestion control.

- ✓ Some local radio stations have helicopters flying around their cities to report on road congestion to make it possible for their mobile listeners to route their packets (cars) around hotspots. This is called traffic-aware routing.

- ✓ In a virtual-circuit network, new connections can be refused if they would cause the network to become congested. This is called **admission control**.
- ✓ **Two difficulties** with this approach are how to identify the onset of congestion, and how to inform the source that needs to slow down.
- ✓ **First issue**, routers can monitor the average load, queuing delay, or packet loss. the **second issue**, routers must participate in a feedback loop with the sources.
- ✓ Finally, when all else fails, the network is forced to discard packets that it cannot deliver. The general name for this is load shedding.

2. Traffic-Aware Routing

- ✓ The goal in taking load into account when computing routes is to shift traffic away from hotspots that will be the first places in the network to experience congestion.
- ✓ Consider the network of Fig. 5-23, which is divided into two parts, East and West, connected by two links, CF and EI.

- ✓ Suppose that most of the traffic between East and West is using link CF, and, as a result, this link is heavily loaded with long delays.

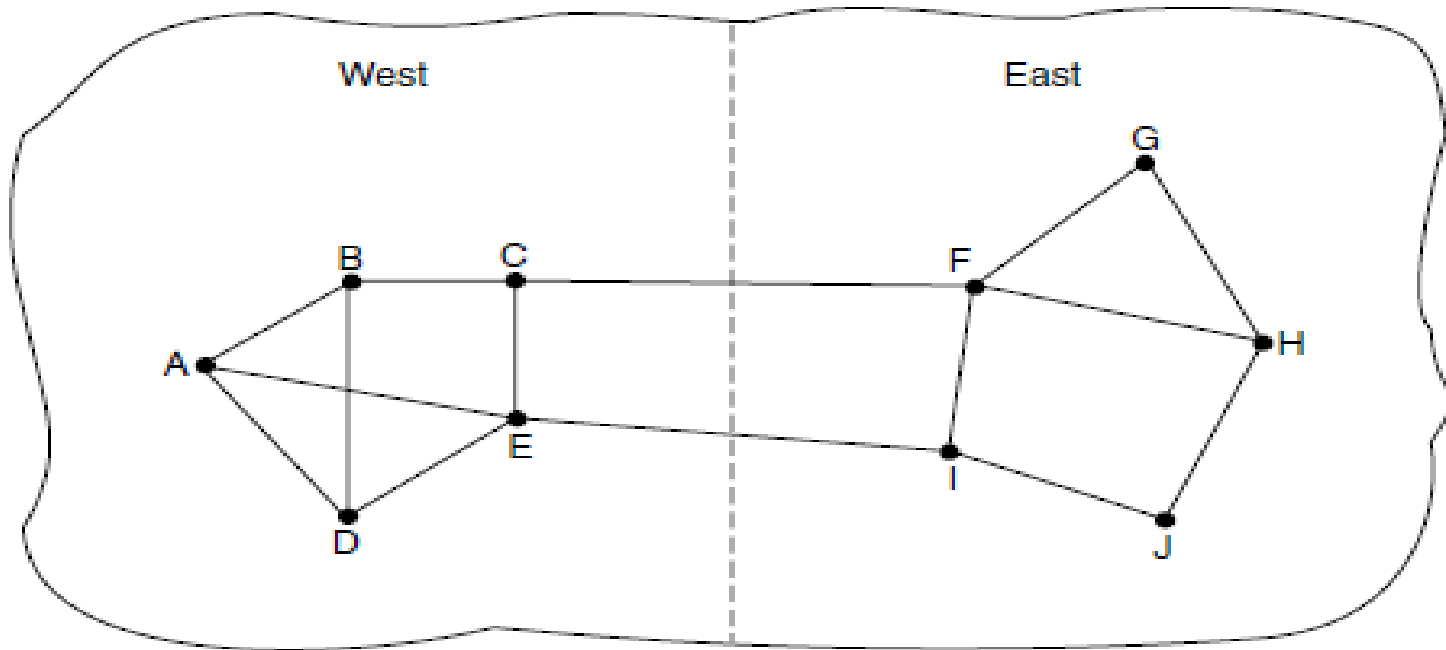


Fig:5-23. A network in which the East and West parts are connected by two links.

- ✓ Two techniques can contribute to a successful solution. The first is multipath routing, in which there can be multiple paths from a source to a destination.
- ✓ In our example this means that the traffic can be spread across both of the East to West links. The second one is for the routing scheme to shift traffic across routes slowly enough that it is able to converge, as in the scheme of Gallagher (1977).

3. Admission Control

- ✓ One technique that is widely used in virtual-circuit networks to keep congestion at bay is admission control.
- ✓ The idea is simple: do not set up a new virtual circuit unless the network can carry the added traffic without becoming congested.
- ✓ The problem of how to describe it in a simple yet meaningful way is difficult because traffic is typically bursty the average rate is only half the story.
- ✓ A commonly used descriptor that captures this effect is the leaky bucket or token bucket(used for quality of service).

- ✓ Admission control can also be combined with traffic-aware routing by considering routes around traffic hotspots as part of the setup procedure.
- ✓ For example, consider the network illustrated in Fig. 5-24(a), in which two routers are congested, as indicated.

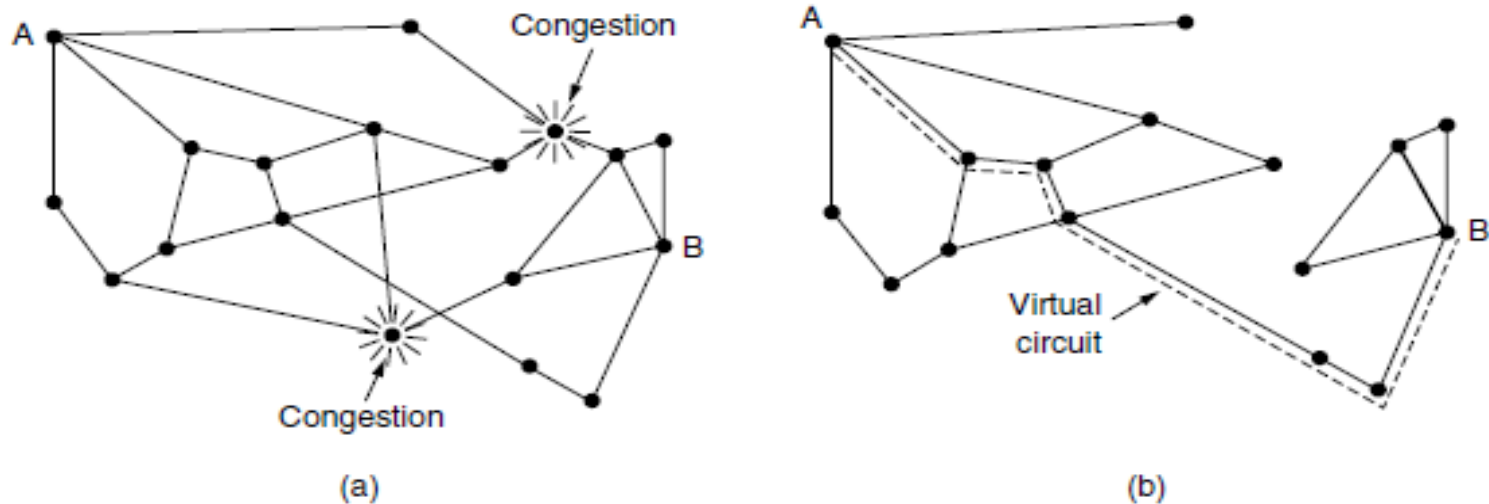


Figure 5-24. (a) A congested network. (b) The portion of the network that is not congested. A virtual circuit from *A* to *B* is also shown.

4. Traffic Throttling

- ✓ The term congestion avoidance is sometimes used to contrast this operating point with the one in which the network has become (overly) congested.
- ✓ Let us now look at some approaches to throttling traffic that can be used in both datagram networks and virtual-circuit networks. Each approach must solve two problems.

Choke Packets

- ✓ In this approach, the router selects a congested packet and sends a choke packet back to the source host, giving it the destination found in the packet.
- ✓ An example of a choke packet used in the early Internet is the SOURCEQUENCH message.
- ✓ It never caught on, though, partly because the circumstances in which it was generated and the effect it had were not clearly specified.

Explicit Congestion Notification

- ✓ This design is called ECN (Explicit Congestion Notification) and is used in the Internet. It is a refinement of early congestion signaling protocols.
- ✓ Two bits in the IP packet header are used to record whether the packet has experienced congestion. Packets are unmarked when they are sent, as illustrated in Fig. 5-25.

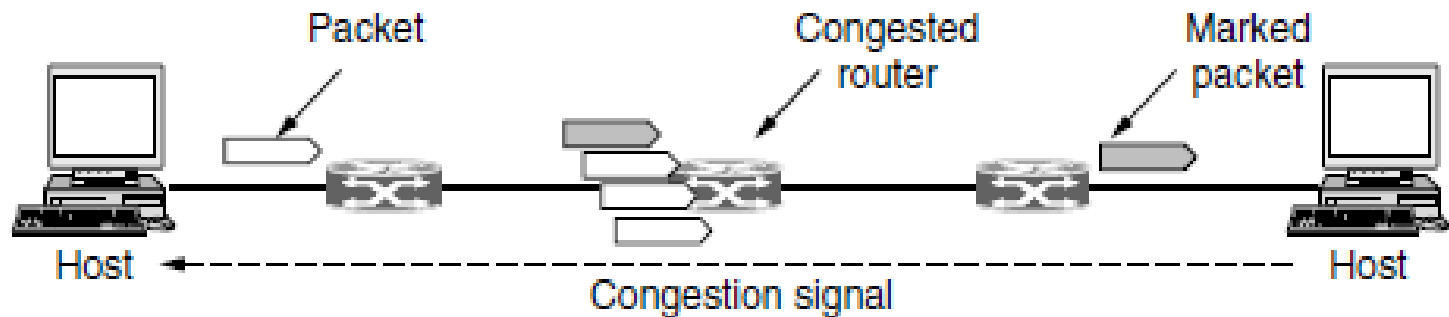


Figure 5-25. Explicit congestion notification

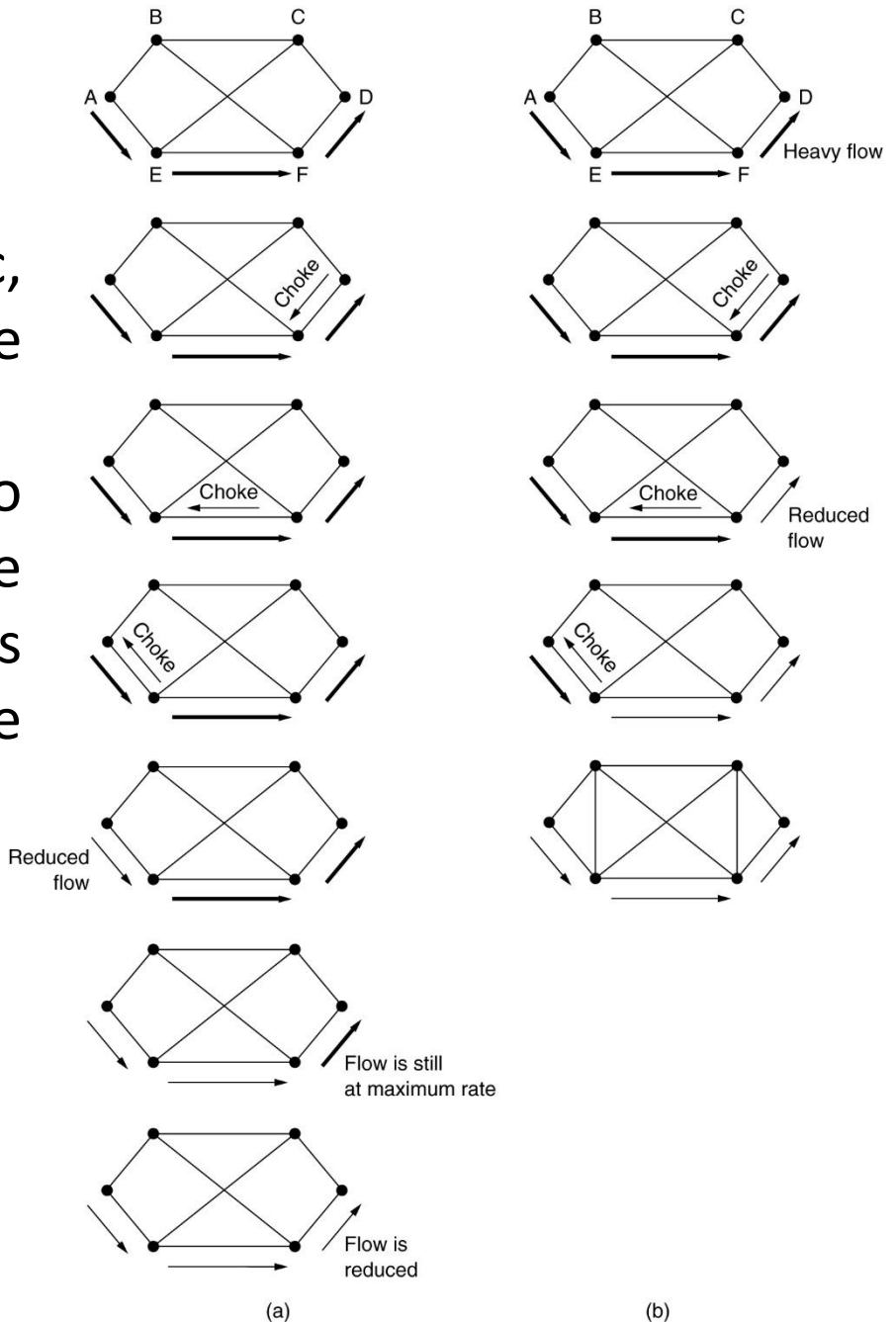
Hop-by-Hop Backpressure

- ✓ Many new packets may be transmitted after congestion has been signaled because of the delay before the signal takes effect. Consider, for example, a host in San Francisco (router A in Fig. 5-26.)

Hop-by-Hop Choke Packets

- ✓ Fig. 5-26(a). In those 40 msec, another 6.2 megabits will have been sent.
- ✓ An alternative approach is to have the choke packet take effect at every hop it passes through, as shown in the sequence of Fig. 5-26(b).

- (a) A choke packet that affects only the source.
- (b) A choke packet that affects each hop it passes through.



Load Shedding

- **Load shedding** is a fancy way of saying that when routers are being inundated by packets that they **cannot handle, they just throw them away.**
- The term comes from the **world of electrical power generation**, where it refers to the practice of utilities intentionally **blacking out certain areas to save the entire grid from collapsing on hot summer days** **when the demand for electricity greatly exceeds the supply.**
- The key question for a router drowning in packets is which packets to drop.
- The preferred choice may depend on the type of applications that use the network.
- For a **file transfer**, an **old packet is worth more than a new one.**
- Real-time media, a new packet is worth more than an old one.
- **Wine & Milk**

- To implement an **intelligent discard policy**, applications must **mark their packets** to indicate to the network **how important they are**.
- Then, when packets have to be discarded, routers can first **drop packets from the least important class**, then the next most important class, and so on.
- For example, the network might let senders send faster than the service they purchased allows if they **mark excess packets as low priority**.
- Such a strategy is actually not a bad idea because it makes **more efficient use of idle resources**, allowing hosts to use them as long as nobody else is interested, but without establishing a right to them when times get tough.

Jitter Control

- The variation (i.e., standard deviation) in the packet arrival times is called **jitter**.
- The **jitter can be bounded** by computing the expected transit time for each hop along the path.
- When a packet arrives at a router, the router checks to see **how much the packet is behind or ahead** of its schedule.
- This information is stored in the packet and updated at each hop. If the packet is ahead of schedule, it is held just long enough to get it back on schedule. If it is behind schedule, the router tries to get it out the door quickly.

Random Early Detection

- Floyd and Jacobson, 1993
- To discard packets before all the buffer space is really exhausted.
- Instead, the only reliable indication of congestion that hosts get from the network is packet loss. After all, it is difficult to build a router that does not drop packets when it is overloaded.
- To determine when to start discarding, routers maintain a running average of their queue lengths. When the average queue length on some link exceeds a threshold, the link is said to be congested and a small fraction of the packets are dropped at random.
- The affected sender will notice the loss when there is no acknowledgement, and then the transport protocol will slow down.

- The lost packet is thus delivering the same message as a choke packet, but implicitly, without the router sending any explicit signal.
- RED routers improve performance compared to routers that drop packets only when their buffers are full, though they may require tuning to work well.
- RED is used when hosts cannot receive explicit signals.

The Network Layer Principles

1. Make sure it works
2. Keep it simple
3. Make clear choices
4. Exploit modularity
5. Expect heterogeneity
6. Avoid static options and parameters
7. Look for good design (not perfect)
8. Strict sending, tolerant receiving
9. Think about scalability
10. Consider performance and cost

1. Make sure it works.

Do not finalize the design or standard until multiple prototypes have successfully communicated with each other.

2. Keep it simple.

When in doubt, use the simplest solution.

If a feature is not absolutely essential, leave it out, especially if the same effect can be achieved by combining other features.

3. Make clear choices.

If there are several ways of doing the same thing, choose one.

4. Exploit modularity.

This principle leads directly to the idea of having protocol stacks, each of whose layers is independent of all the other ones.

5. Expect heterogeneity.

Different types of hardware, transmission facilities, and applications will occur on any large network. To handle them, the network design must be simple, general, and flexible.

6. Avoid static options and parameters.

If parameters are unavoidable (e.g., maximum packet size), it is best to have the sender and receiver negotiate a value than defining fixed choices.

7. Look for a good design; it need not be perfect.

Often the designers have a good design but it cannot handle some weird special case.

8. Be strict when sending and tolerant when receiving.

In other words, only send packets that rigorously comply with the standards, but expect incoming packets that may not be fully conformant and try to deal with them.

9. Think about scalability.

If the system is to handle millions of hosts and billions of users effectively, no centralized databases of any kind are tolerable and load must be spread as evenly as possible over the available resources.

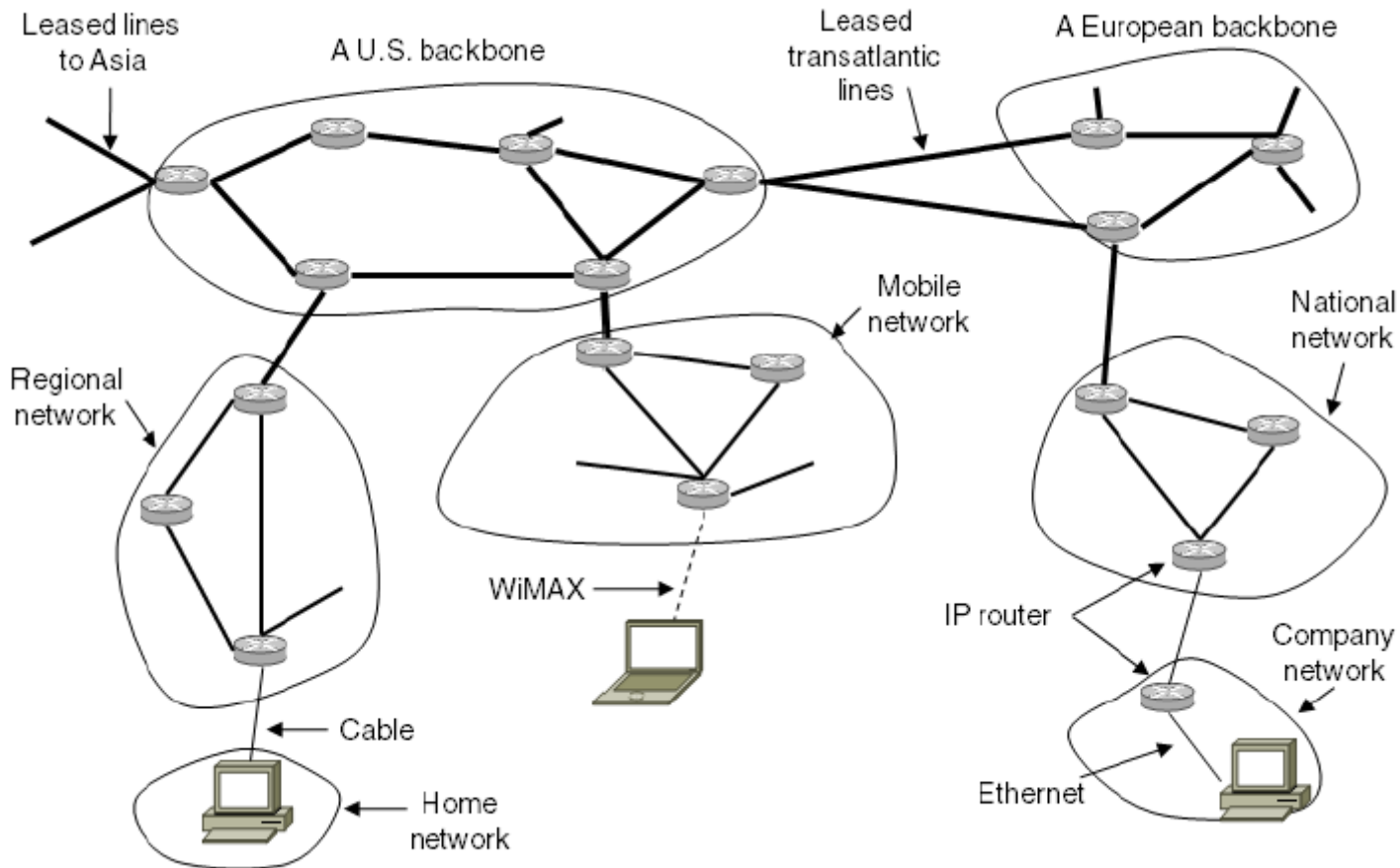
10. Consider performance and cost.

If a network has poor performance or outrageous costs, nobody will use it.

The Network Layer in the Internet

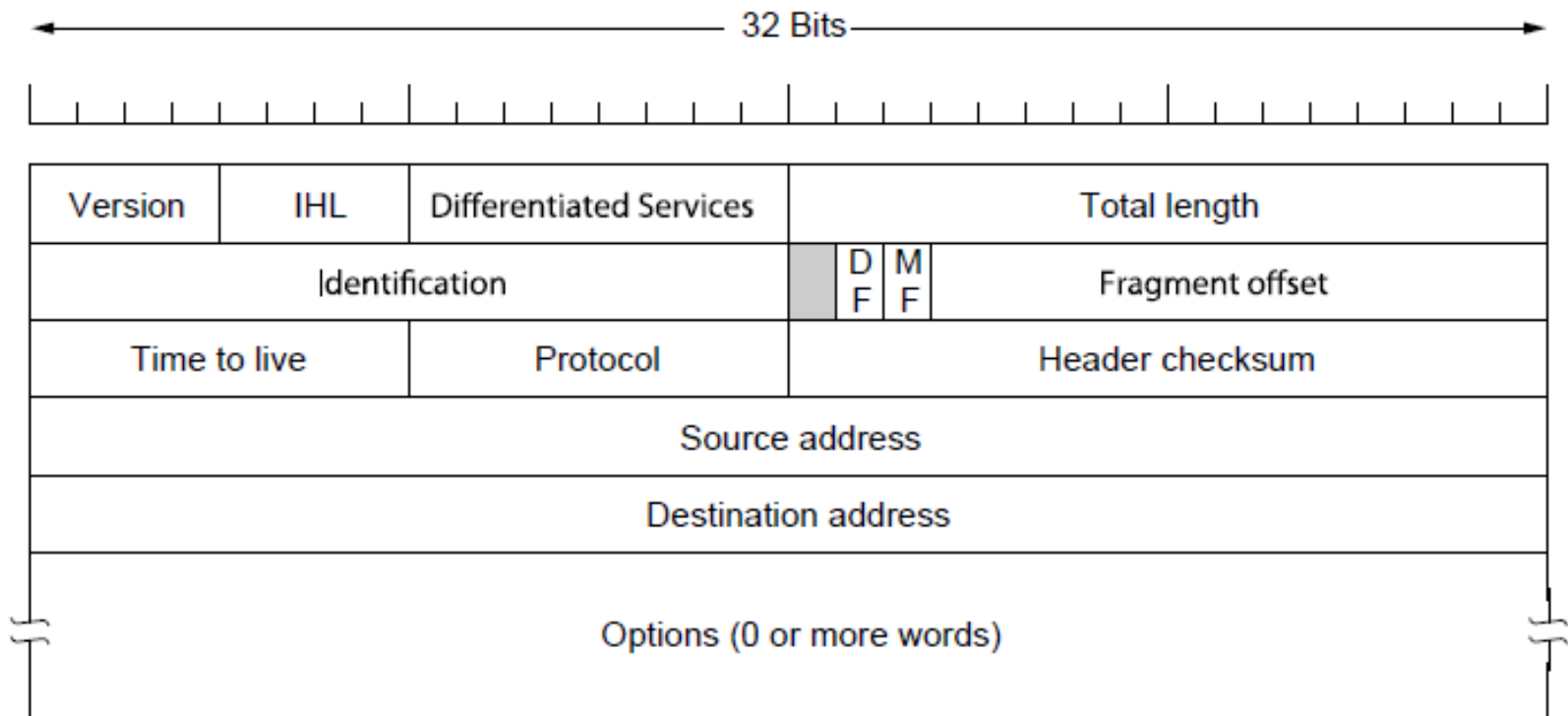
- The IP Version 4 Protocol
- IP Addresses
- IP Version 6
- Internet Control Protocols
- Label Switching and MPLS
- OSPF (Open Shortest Path First)— An Interior Gateway Routing Protocol
- BGP—The Exterior Gateway Routing Protocol
- Internet Multicasting
- Mobile IP

The Network Layer in the Internet



The Internet is an interconnected collection of many networks.

The IP Version 4 Protocol header.



- The **Differentiated services** field or **Type of service** field.
- The **Type of service** field provided 3 bits to signal priority and 3 bits to signal whether a host cared more about delay, throughput, or reliability.
 - The header has a 20-byte fixed part and a variable-length optional part.
 - The bits are transmitted from left to right and top to bottom, with the high-order bit of the *Version* field going first. (This is a **“big-endian” network byte order.**)
 - The **header length** is not constant, a field in the header, **IHL**, is provided to tell how long the header is, in 32-bit words.

- The *Total length* includes everything in the datagram—both header and data. **The maximum length is 65,535 bytes.**
- The *Identification field* is needed to allow the destination host to determine which packet a newly arrived fragment belongs to. All the fragments of a packet contain the same *Identification* value.
- *DF* stands for **Don't Fragment**. It is an order to the routers not to fragment the packet.
- *MF* stands for **More Fragments**. All fragments except the last one have this bit set. It is needed to know when all fragments of a datagram have arrived.
- The *Fragment offset* tells where in the current packet this fragment belongs. All fragments except the last one in a datagram must be a multiple of 8 bytes, the elementary fragment unit.
- The *TtL (Time to live)* field is a counter used to **limit packet lifetimes**. It was originally supposed to count time in seconds, allowing a **maximum lifetime of 255 sec.**
- The *Protocol field* tells it which transport process to give the packet to. TCP is one possibility, but so are UDP and some others.

- **Header checksum** is useful for **detecting errors** while the packet travels through the network. Note that it must be recomputed at each hop because at least one field always changes (the *Time to live* field), but tricks can be used to speed up the computation.
- The *Source address* and *Destination address* indicate the IP address of the source and destination network interfaces. We will discuss Internet addresses in the next section.
- The *Options field* was designed to provide an escape to allow subsequent versions of the protocol to include information not present in the original design, to permit experimenters to try out new ideas, and to avoid allocating header bits to information that is rarely needed.
- The *Options field* is padded out to a multiple of 4 bytes.

The IP Version 4 Protocol

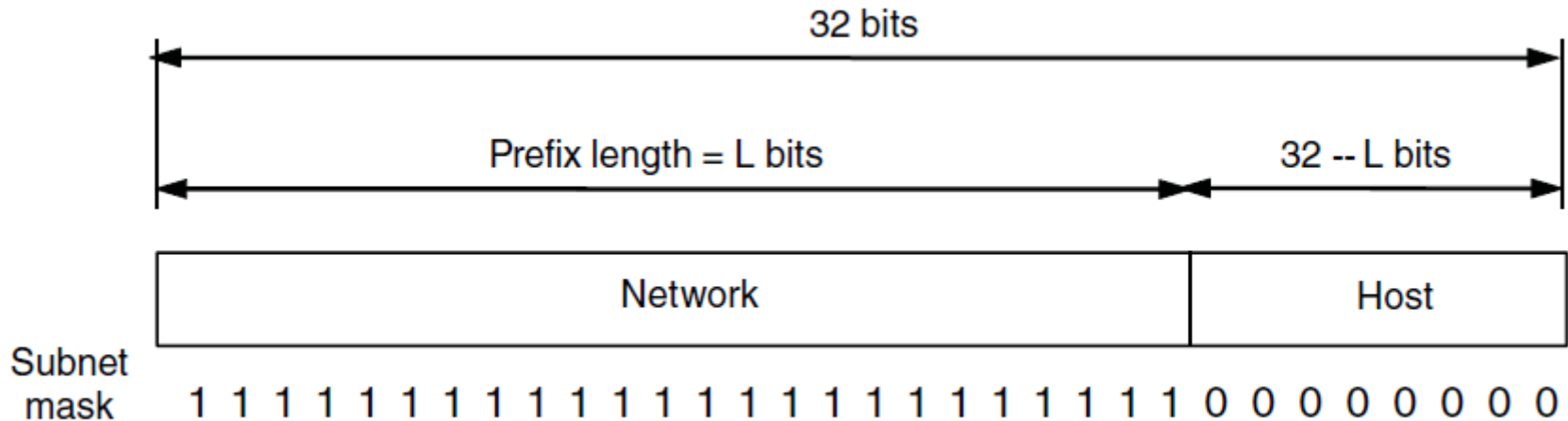
- The **minimum value** is **5**, which applies when **no options are present**.
- The maximum value of this 4-bit field is 15, which limits the header to 60 bytes, and thus the *Options* field to 40 bytes.

Option	Description
Security	Specifies how secret the datagram is
Strict source routing	Gives the complete path to be followed
Loose source routing	Gives a list of routers not to be missed
Record route	Makes each router append its IP address
Timestamp	Makes each router append its address and timestamp

Some of the IP options.

IP Addresses :An IP prefix.

The subnet mask is 255.255.255.0. shows a prefix and a subnet mask.



- A defining feature of IPv4 is its 32-bit addresses.
- It is important to note that an IP address does not actually refer to a host. It really refers to a network interface, so if a host is on two networks, it must have two IP addresses.
- one network have one IP address. In contrast, routers have multiple interfaces and thus multiple IP addresses.
- IP addresses are written in **dotted decimal notation**. Each of the 4 bytes is written in decimal, from 0 to 255.
- Prefixes are simply described by their length, as in a “/16”.
- The length of the prefix corresponds to a binary mask of 1s in the network portion, it is called a **subnet mask**.

2. IP Addresses

- ✓ Every host and router on the Internet has an IP address that can be used in the Source address and Destination address fields of IP packets. It is important to note that an IP address does not actually refer to a host.

Prefixes

- ✓ IP addresses are hierarchical, unlike Ethernet addresses. The network portion has the same value for all hosts on a single network, such as an Ethernet LAN. This means that a network corresponds to a contiguous block of IP address space. This block is called a prefix.
- ✓ IP addresses are written in dotted decimal notation. In this format, each of the 4 bytes is written in decimal, from 0 to 255. For example, the 32-bit hexadecimal address 80D00297 is written as 128.208.2.151.
- ✓ Prefixes are written by giving the lowest IP address in the block and the size of the block.
- ✓ The length of the prefix corresponds to a binary **mask of 1s** in the network portion. When written out this way, it is called a **subnet mask**.

- The **key advantage of prefixes** is that routers can forward packets based on only the network portion of the address, as long as each of the networks has a unique address block.
- While using a hierarchy lets Internet routing scale, it has **two disadvantages**.
- **First**, the IP address of a host depends on where it is located in the network.
- **The second disadvantage** is that the hierarchy is wasteful of addresses unless it is carefully managed.

Subnets

- Network numbers are managed by a nonprofit corporation called **ICANN (Internet Corporation for Assigned Names and Numbers)**, to avoid conflicts.
- IP address assignment is ongoing as companies grow.
- The solution is to allow the block of addresses to be split into several parts for internal use as multiple networks, while still acting like a single network to the outside world. This is called **subnetting** and the networks (such as Ethernet LANs) that result from **dividing up a larger network** are called **subnets**.

Subnetwork, or **subnet**, is a logically visible, distinctly addressed part of a single Internet Protocol network.

Subnetting is the division of a computer network into groups of computers that have a common, designated IP address routing prefix.

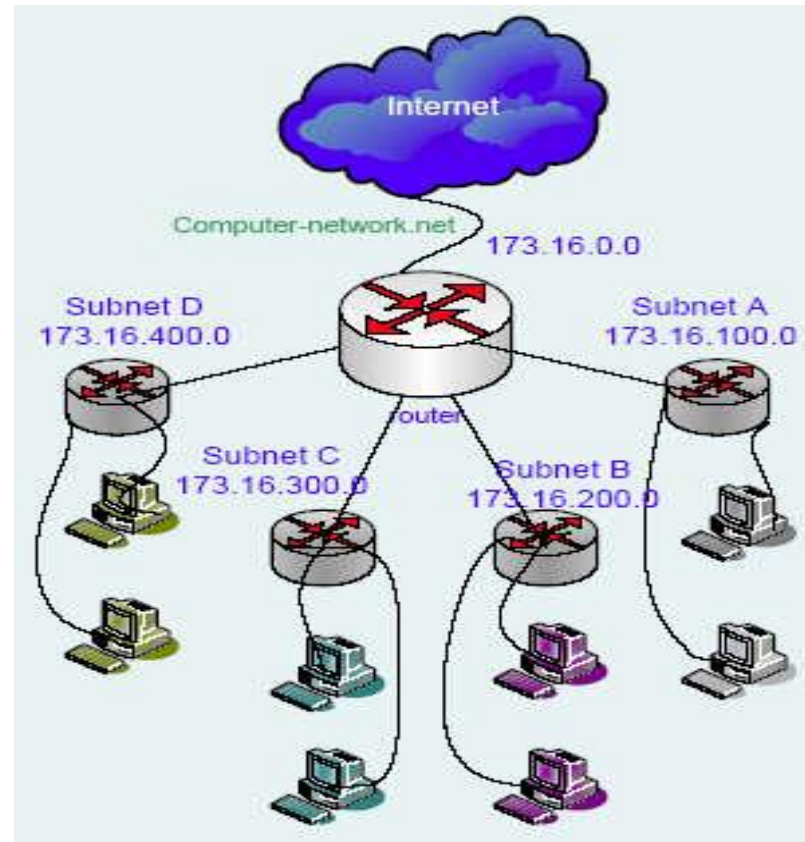
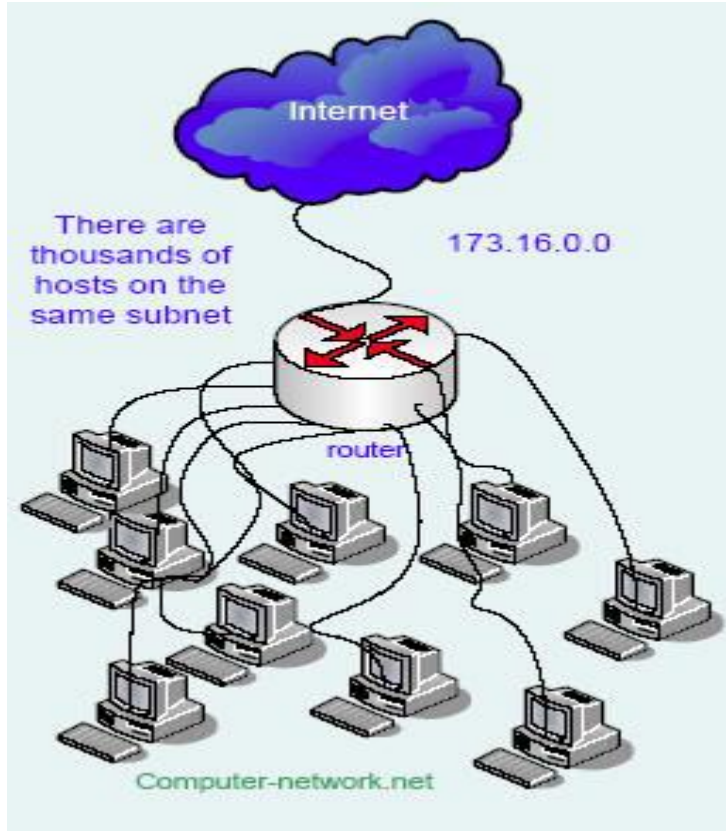
Advantages

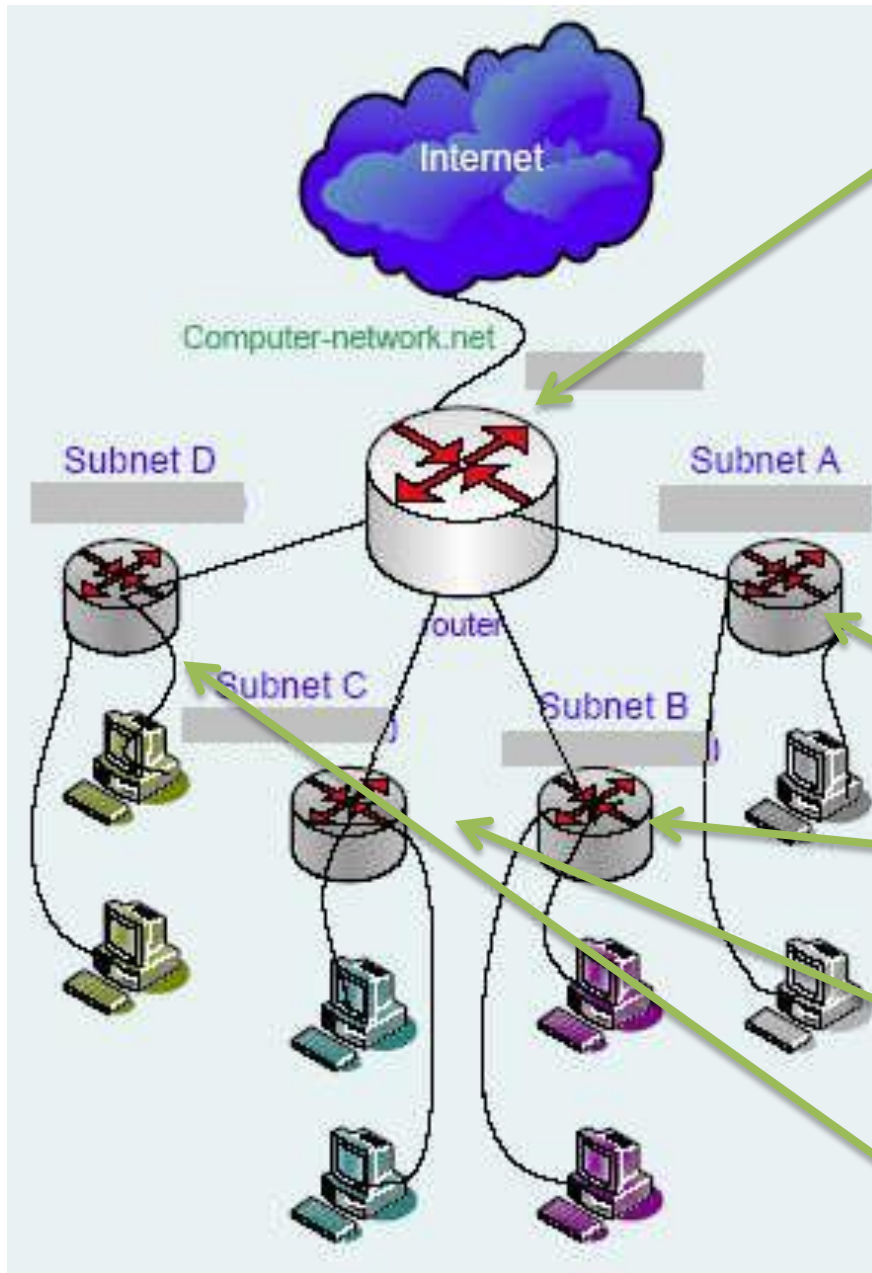
- Allows a single shared network address to split it up into many smaller networks.
- Without subnets, organizations would require many network addresses
 - Limited number of Network addresses available
- **Traffic**
 - Smaller routing tables
 - Alleviates excessive packet collision and congestion
- Easier to manage and solve problems
- **Better Security**
 - Separating departments with highly sensitive material
 - Accounting and Administration

Disadvantages

- Doesn't allocate IP address proportionately per subnet
- Limited by the number of IP address
- Need to buy hardware such as routers

Subnet Applications





192.168.5.130 /24

Subnet mask = 255.255.255.0

192.168.5.0 = Network ID

4 Subnets = 62 hosts/subnet

Borrow 2 bits from host byte

Subnet Mask = 255.255.255.192
= /26

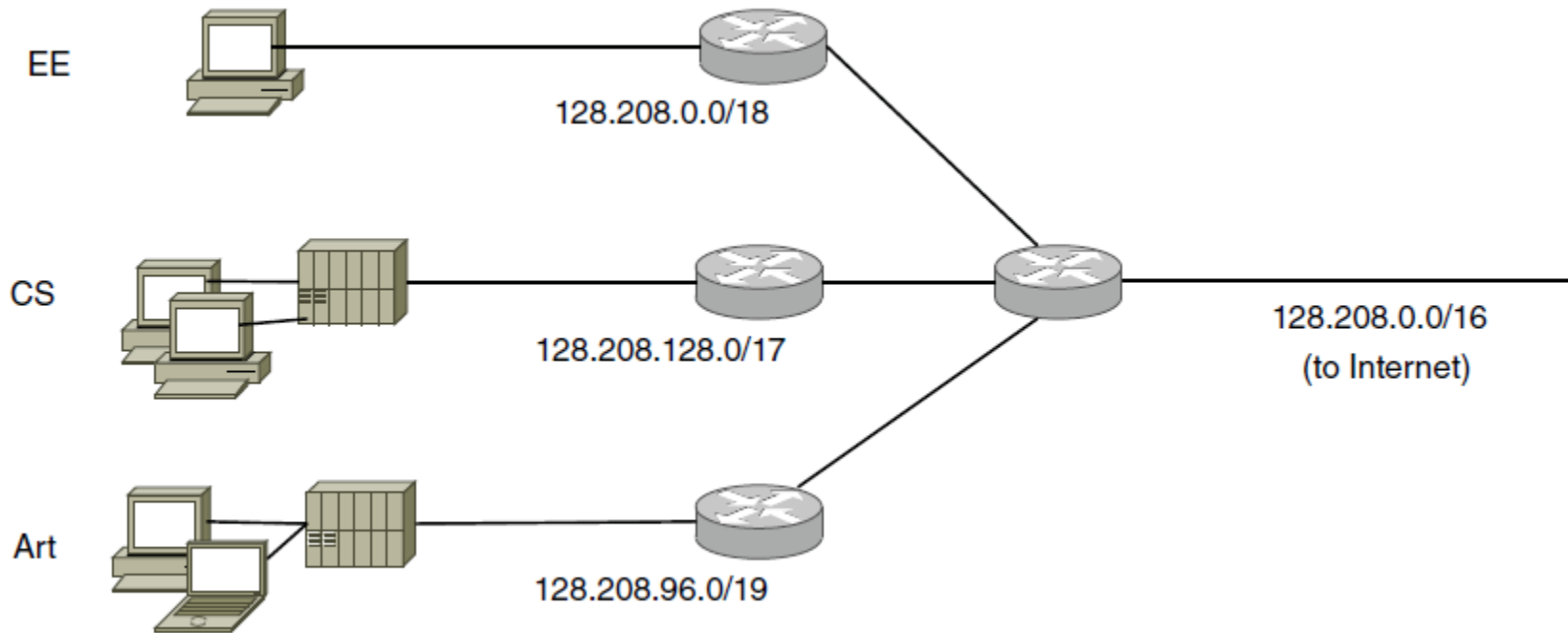
Subnet A -> 192.168.5.1/26
to 192.168.5.62/26

Subnet B -> 192.168.5.65/26
to 192.168.5.126/26

Subnet C -> 192.168.5.129/26
to 192.168.5.190/26

Subnet D -> 192.168.5.193/26
to 192.168.5.254/26

IP Addresses (2)



Splitting an IP prefix into separate networks with subnetting.

- The vertical bar (|) shows the boundary between the subnet number and the host portion.
- Outside the network, the subnetting is not visible, so allocating a new subnet does not require contacting ICANN or changing any external databases.

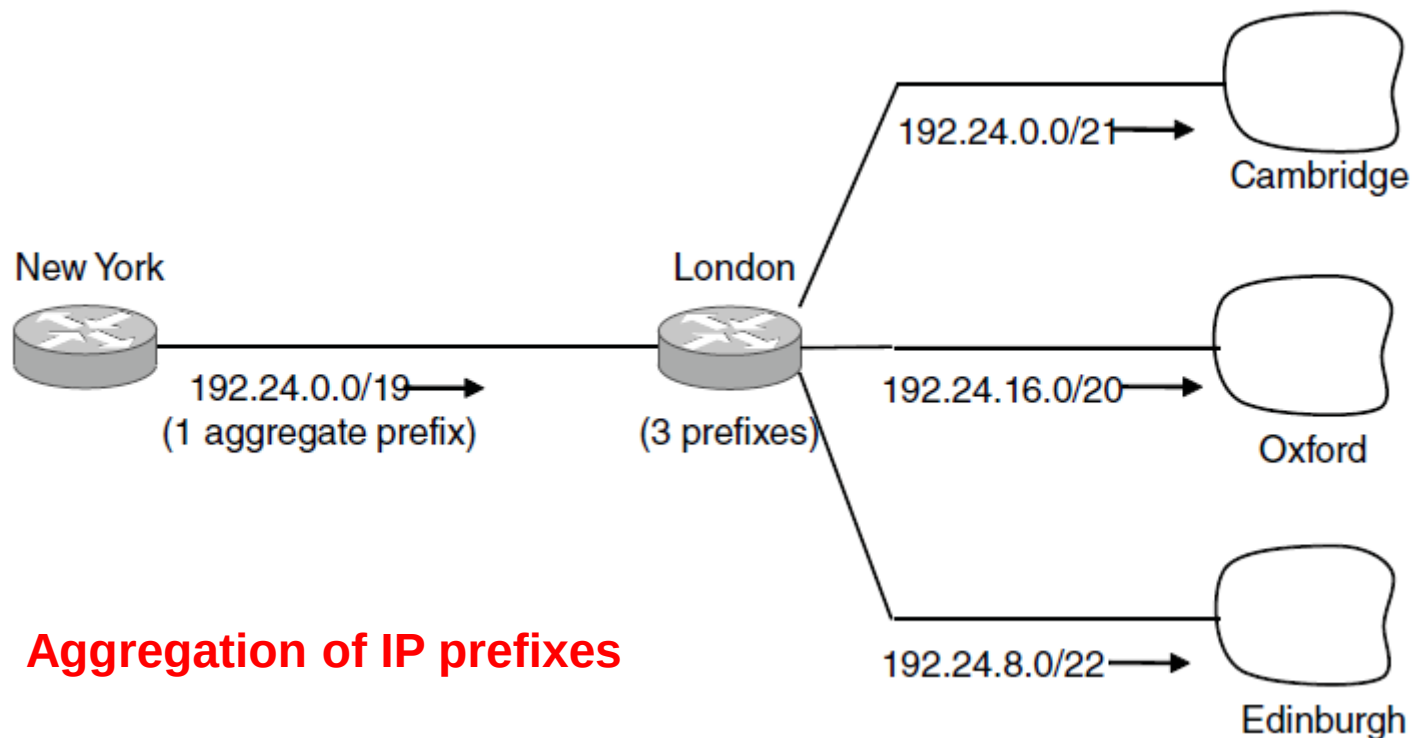
- When a packet arrives, the router looks at the destination address of the packet and checks which subnet it belongs to.
- Destination IP address is 128.208.2.151.
- To see if it is for the Computer Science Dept., we AND with 255.255.128.0 to take the first 17 bits (which is 128.208.0.0) and see if they match the prefix address of CS dept. (which is 128.208.128.0).

CIDR—Classless Inter Domain Routing

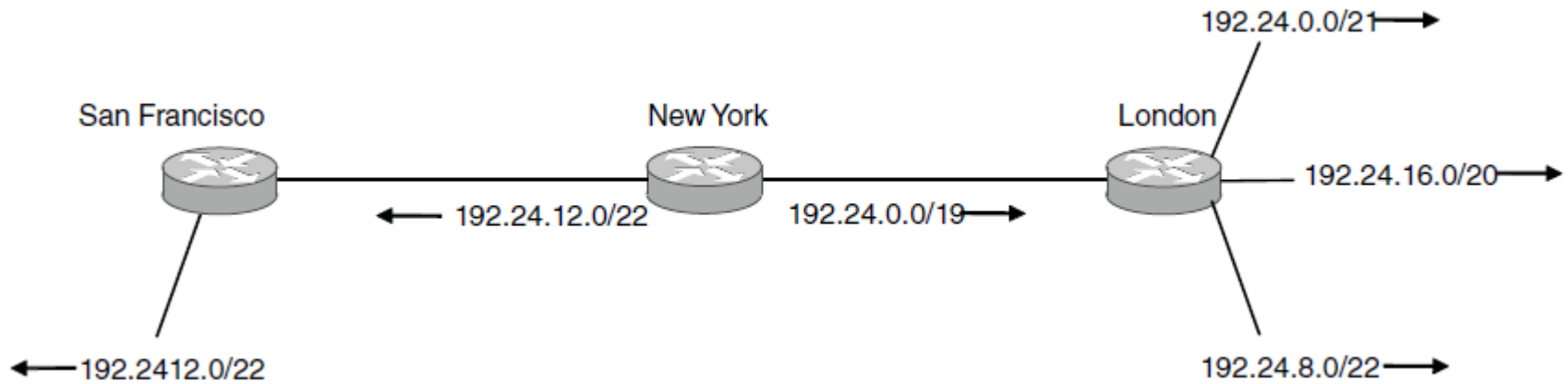
- Even if blocks of IP addresses are allocated so that the addresses are used efficiently, there is still a **problem** that remains: **routing table explosion**.
- Routers in ISPs and backbones must know which way to go to get to every network and no simple default will work. These core routers are said to be in the **default-free zone** of the Internet.
- **To reduce routing table sizes.** We can apply the same insight as subnetting: routers at different locations can know about a given IP address as belonging to prefixes of different sizes. However, **instead of splitting an address block into subnets**, here we combine multiple small prefixes into a single larger prefix. This process is **called route aggregation**.
- The resulting **larger prefix** is sometimes called a **supernet**, to contrast with subnets as the **division of blocks of addresses**.
- Each router to have the corresponding prefix information and works with subnetting, is called **CIDR (Classless Inter- Domain Routing)**.

IP Addresses A set of IP address assignments

University	First address	Last address	How many	Prefix
Cambridge	194.24.0.0	194.24.7.255	2048	194.24.0.0/21
Edinburgh	194.24.8.0	194.24.11.255	1024	194.24.8.0/22
(Available)	194.24.12.0	194.24.15.255	1024	194.24.12.0/22
Oxford	194.24.16.0	194.24.31.255	4096	194.24.16.0/20



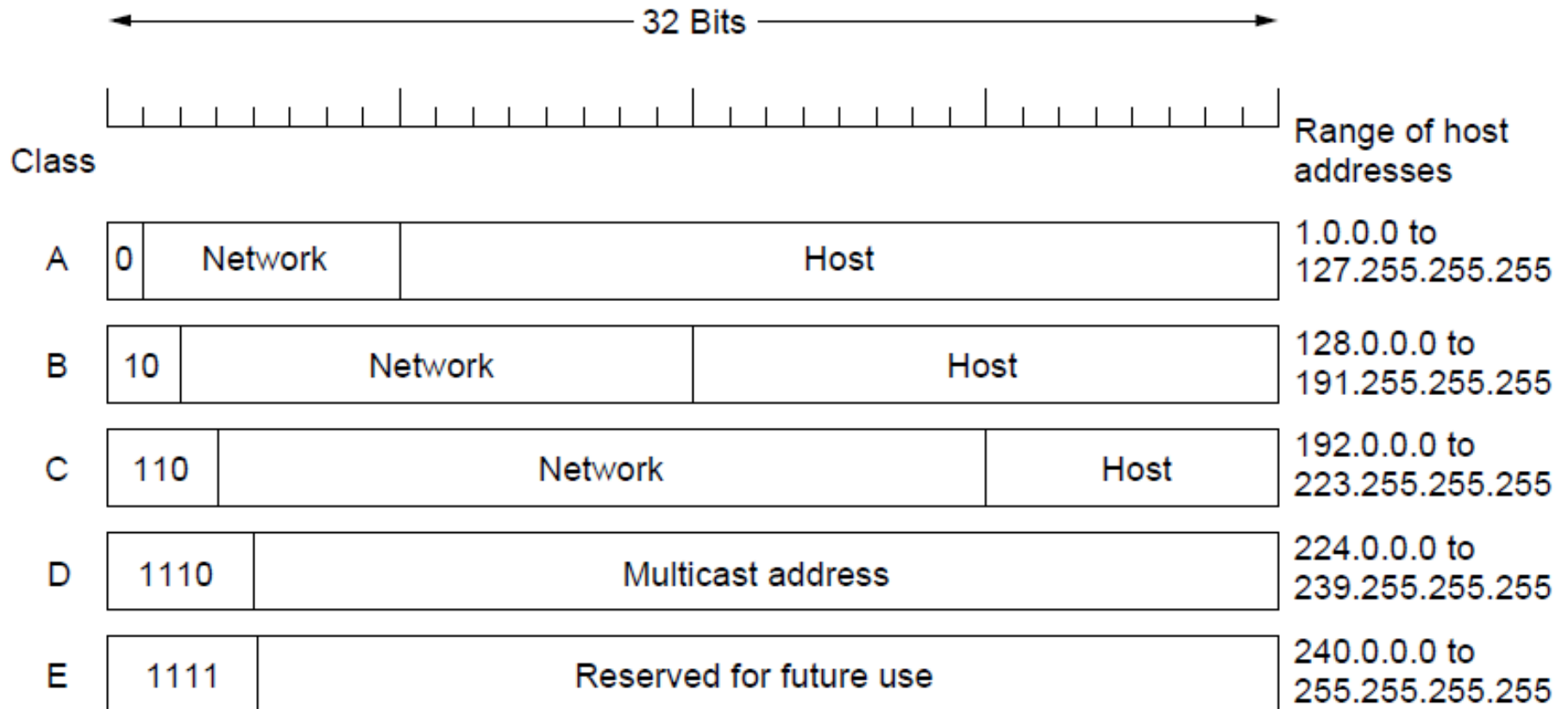
- Conceptually, **CIDR** works as follows. When a packet comes in, the routing table is scanned to determine if the destination lies within the prefix.
- It is possible that **multiple entries** with **different prefix lengths** will **match**, in which case the entry with the **longest prefix** is **used**.



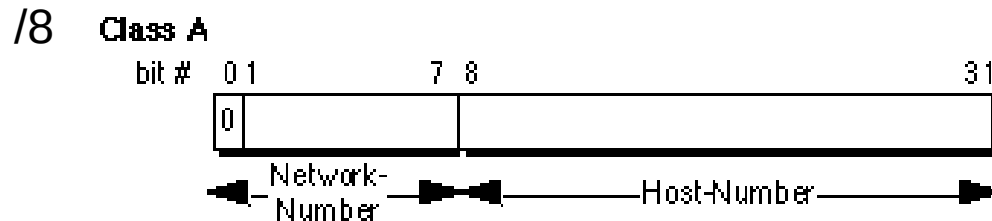
Longest matching prefix routing at the New York router.

Instead, complex algorithms have been devised to speed up the address matching process (Ruiz-Sanchez et al., 2001). Commercial routers use custom VLSI chips with these algorithms embedded in hardware.

Classful and Special Addressing

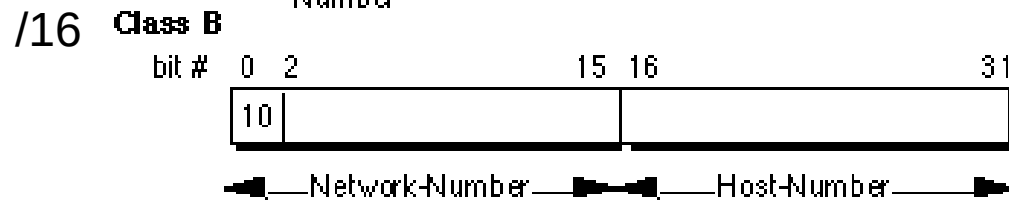


IP address formats



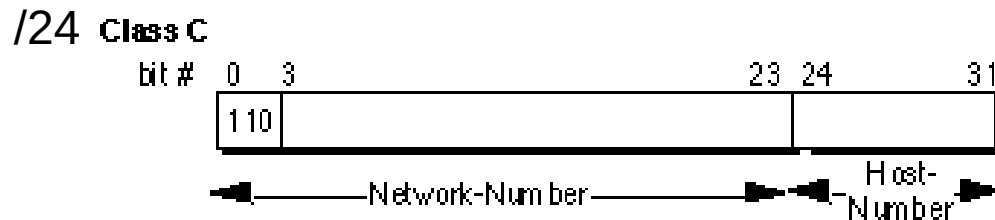
$$2^7 - 2 = 126 \text{ networks}$$

$$2^{24} - 2 = 16,777,214 \text{ hosts / network}$$



$$2^{14} = 16,384 \text{ networks}$$

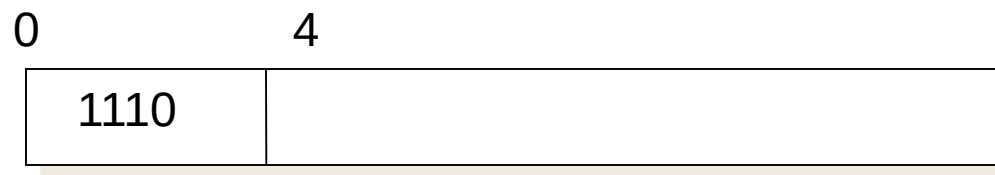
$$2^{16} - 2 = 65,534 \text{ hosts / network}$$



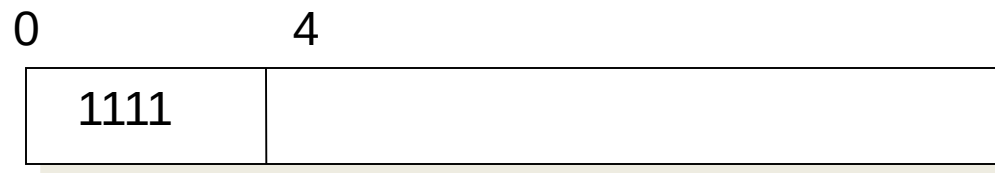
$$2^{21} = 2,097,152 \text{ networks}$$

$$2^8 - 2 = 254 \text{ hosts / network}$$

Class D: (IP Multicasting)



Class E: (Experimental use)



IP Address Class Assignments	
Class	First Octet Value
Class A	0 ~ 127
Class B	128 ~ 191
Class C	192 ~ 223
Class D	224 ~ 239
Class E	240 ~ 255

- The **number of hosts** available in any of the class addresses, where “n” represents the number of bits in the host portion:

$$(2^n - 2) = \text{Number of available hosts}$$

- For a Class A network, there are:

$$2^{24} - 2 \text{ or } 16,777,214 \text{ hosts.}$$

- Class A address uses 7 bits to designate the network, so

$$(2^7 - 2) = 126 \text{ or there can be } 126 \text{ Class A Networks.}$$

- Class B ($2^{14} - 2$), there can be **16,382** Class B Networks & each Network can have ($2^{16} - 2$) Hosts, or **65,534** Hosts.

- The class A, B, and C formats allow for up to **128 networks** with **16 million hosts** each, **16,384 networks** with up to **65,536 hosts** each, and **2 million networks** (e.g., LANs) with up to 256 hosts each (although a few of these are special).
- Also supported is multicast (the class D format), in which a datagram is directed to multiple hosts. Addresses beginning with 1111 are reserved for use in the future.
- **Class A** network, with **16 million addresses**, is too big, and a **class C network**, with **256 addresses** is too small. A **class B** network, with **65,536**, is just right. This situation is known as the **three bears problem**

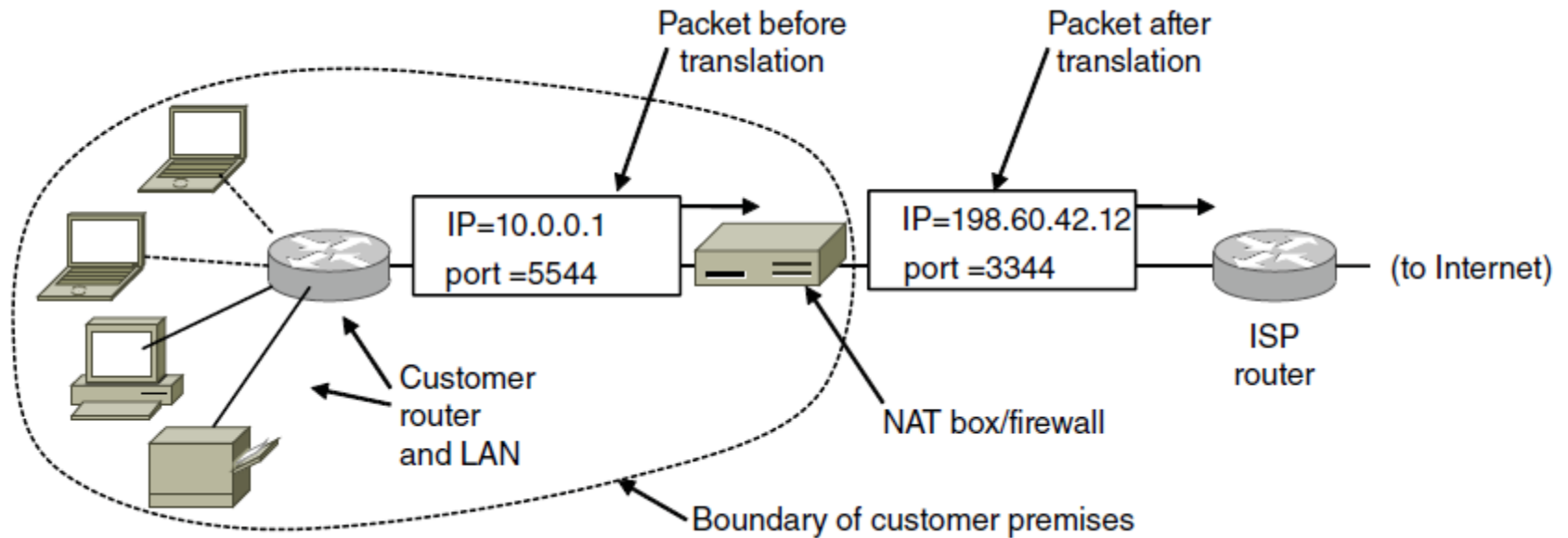
IP Addresses (7)

0 0																														This host										
0 0										...										0 0										Host										A host on this network
1 1																														Broadcast on the local network										
Network										1 1 1 1										...										1 1 1 1										Broadcast on a distant network
127										(Anything)																				Loopback										

Special IP addresses

Default Subnet Masks	
Address Class	Subnet Mask
Class A	255.0.0.0
Class B	255.255.0.0
Class C	255.255.255.0

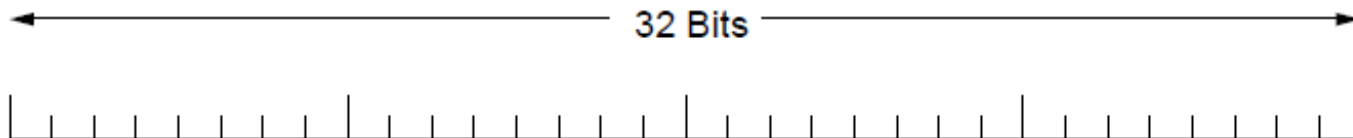
IP Addresses (8)



Placement and operation of a NAT box.

IP Version 6 Goals

- Support billions of hosts
- Reduce routing table size
- Simplify protocol
- Better security
- Attention to type of service
- Aid multicasting
- Roaming host without changing address
- Allow future protocol evolution
- Permit coexistence of old, new protocols. . .



Version	Diff. Serv.	Flow label	
Payload length		Next header	Hop limit
Source address (16 bytes)			
Destination address (16 bytes)			

The IPv6 fixed header (required).

IP Version 6 (2)

Extension header	Description
Hop-by-hop options	Miscellaneous information for routers
Destination options	Additional information for the destination
Routing	Loose list of routers to visit
Fragmentation	Management of datagram fragments
Authentication	Verification of the sender's identity
Encrypted security payload	Information about the encrypted contents

IPv6 extension headers

IPv 6

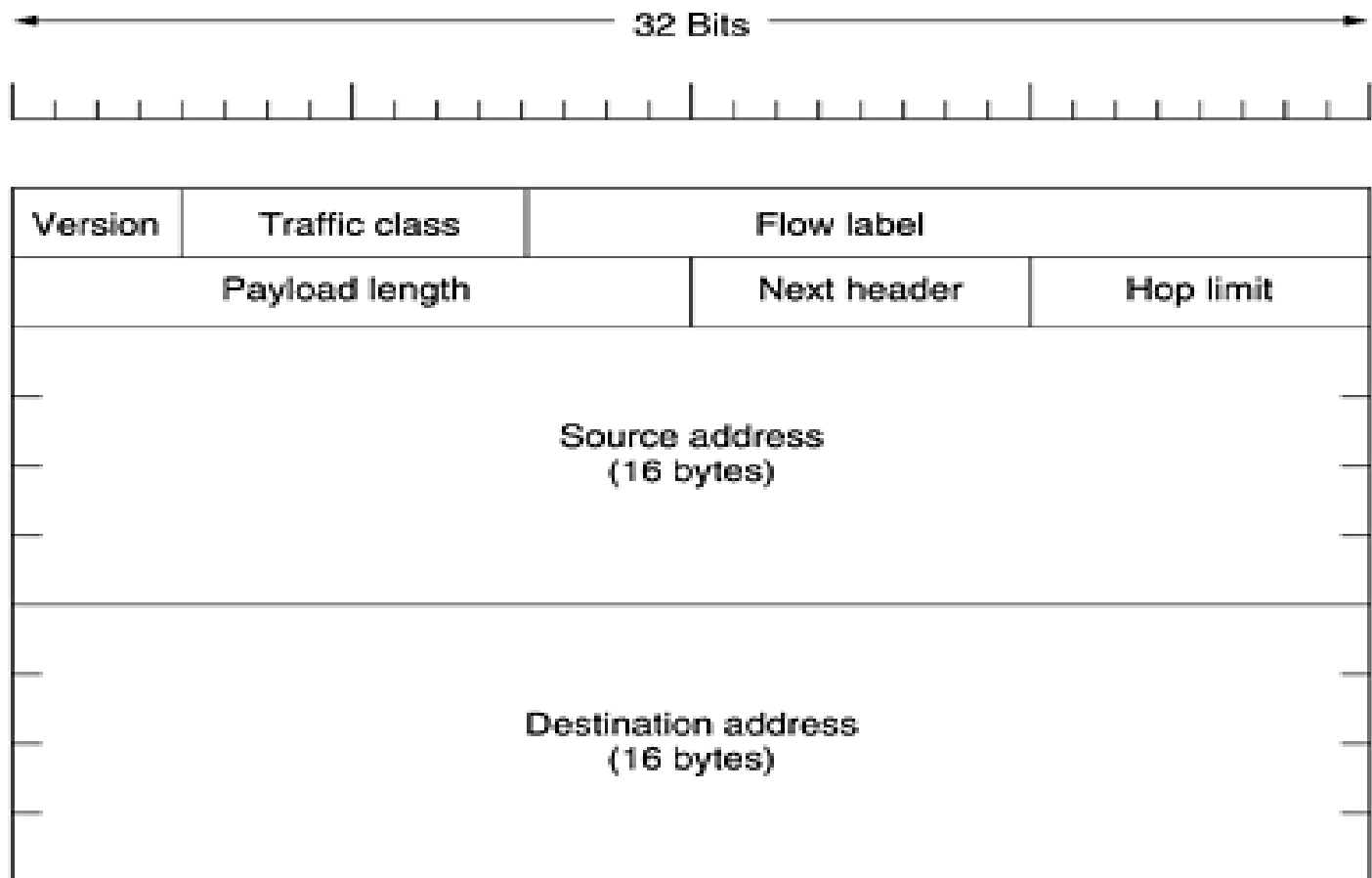
- In 1990, IETF started work on a new version of IP, one which would **never run out of addresses**, would **solve a variety** of other problems, and be **more flexible** and **efficient** as well. Its major goals were:
 - Support billions of hosts, even with inefficient address space allocation.
 - Reduce the size of the routing tables.
 - Simplify the protocol, to allow routers to process packets faster.
 - Provide better security (authentication and privacy) than current IP.
 - Pay more attention to type of service, particularly for real-time data.
 - Aid multicasting by allowing scopes to be specified.
 - Make it possible for a host to roam without changing its address.
 - Allow the protocol to evolve in the future.
 - Permit the old and new protocols to coexist for years.

Features of IPv6

- IPv6 has **longer addresses** than IPv4. They are **16 bytes** long, which solves the problem that IPv6 set out to solve: provide an effectively unlimited supply of Internet addresses.
- **Simplification of the header.** It contains only **seven fields** (versus 13 in IPv4). This change allows routers to process packets faster and thus improve throughput and delay.
- **Better support for options.** This change was essential with the new header because fields that previously were required are now optional. In addition, the way options are represented is different, making it simple for routers to skip over options not intended for them. This feature speeds up packet processing time.
- Big advance is in **security**.
- **More attention has been paid to quality of service.** Various half-hearted efforts have been made in the past, but now with the growth of multimedia on the Internet, the sense of urgency is greater.

The Main IPv6 Header

Figure 5-68. The IPv6 fixed header (required).



- The **Version** field is always 6 for IPv6 (and 4 for IPv4).
- The **Traffic class** field is used to distinguish between packets with different real-time delivery requirements.
- The **Flow label** field is also still experimental but will be used to allow a source and destination to set up a pseudo connection with particular properties and requirements.
 - For example, a stream of packets from one process on a certain source host to a certain process on a certain destination host might have stringent delay requirements and thus need reserved bandwidth.
 - The flow can be set up in advance and given an identifier.
 - When a packet with a **nonzero Flow** label shows up, all the routers can look it up in internal tables to see what kind of special treatment it requires. In effect, flows are an attempt to have it both ways.
 - Each flow is designated by the source address, destination address, and flow number

- The **Payload length** field tells how many bytes follow the 40-byte header.
- The **Next header** field.
 - This field tells which of the (currently) six extension headers, if any, follow this one.
 - If this header is the last IP header, the Next header field tells which transport protocol handler (e.g., TCP, UDP) to pass the packet to.
- The **Hop limit** field is used to keep packets from living forever.

- Next come the **Source address** and **Destination address** fields.
- A new notation has been devised for writing 16-byte addresses. They are written as eight groups of four hexadecimal digits with colons between the groups, like this:
 - **8000:0000:0000:0000:0123:4567:89AB:CDEF**
 - Since many addresses will have many zeros inside them, **three** optimizations have been authorized.
 - First, leading zeros within a group can be omitted, so 0123 can be written as 123.
 - Second, one or more groups of 16 zero bits can be replaced by a pair of colons.
 - Thus, the above address now becomes
 - **8000::123:4567:89AB:CDEF**
 - Finally, IPv4 addresses can be written as a pair of colons and an old dotted decimal number, for example
 - **::192.31.20.46**
- Thus, the goal of IPv6-a **fast, yet flexible, protocol** with plenty of address space-has been met by this design.

Extension Headers

- Six kinds of extension headers are defined at present, as listed in [Fig. 5-69](#).
- Each one is optional, but if more than one is present, they must appear directly after the fixed header, and preferably in the order listed.

Figure 5-69. IPv6 extension headers.

Extension header	Description
Hop-by-hop options	Miscellaneous information for routers
Destination options	Additional information for the destination
Routing	Loose list of routers to visit
Fragmentation	Management of datagram fragments
Authentication	Verification of the sender's identity
Encrypted security payload	Information about the encrypted contents

- **Some** of the headers have a **fixed format**; others contain a variable number of variable-length fields.
- For these, each item is encoded as a (**Type, Length, Value**) tuple.
- The **Type** is a 1-byte field telling which option this is.
- The Type values have been chosen so that the first 2 bits tell routers that do not know how to process the option what to do.
 - The choices are: **skip the option; discard the packet; discard the packet and send back an ICMP packet;**
- The **Length** is also a 1-byte field. It tells how long the value is (0 to 255 bytes).
- The **Value** is any information required, up to 255 bytes.

- The **hop-by-hop header** is used for information that all routers along the path must examine. So far, one option has been defined: support of datagrams exceeding 64K. The format of this header is shown in [Fig. 5-70](#). When it is used, the Payload length field in the fixed header is set to zero.

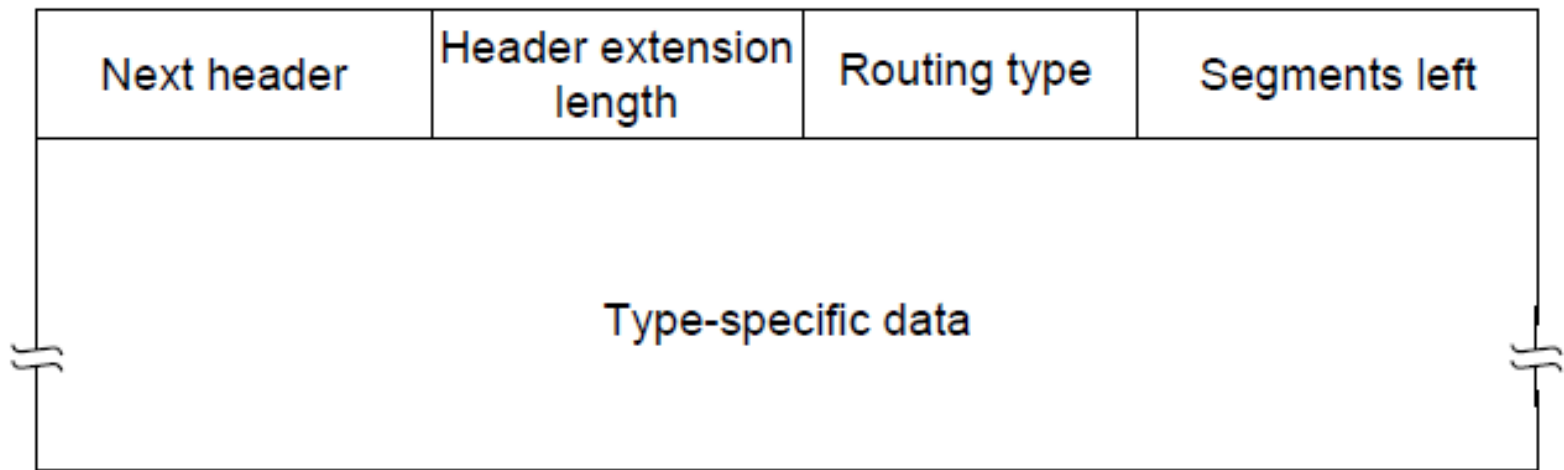
Next header	0	194	4
Jumbo payload length			

The hop-by-hop extension header for large datagrams (jumbograms).

- As with all extension headers, this one starts out with a **byte** telling what kind of header comes next.
- This byte is followed by one telling how long the hop-by-hop header is in bytes, excluding the first 8 bytes.
- The next 2 bytes indicate that this option defines the datagram size (code 194) and that the size is a 4-byte number.
- The last 4 bytes give the size of the datagram. Sizes less than 65,536 bytes are not permitted and will result in the first router discarding the packet and sending back an ICMP error message.
- Datagrams using this header extension are called **jumbograms**.
- The use of **jumbograms** is important for supercomputer applications that must transfer gigabytes of data efficiently across the Internet.

- The **destination options header** is intended for fields that need only be interpreted at the destination host.
- In the initial version of IPv6, the only options defined are null options for padding this header out to a multiple of 8 bytes, so initially it will not be used.
- It was included to make sure that new routing and host software can handle it, in case someone thinks of a destination option some day.

- The **routing header** lists one or more routers that must be visited on the way to the destination.
- It is very similar to the IPv4 loose source routing in that all addresses listed must be visited in order, but other routers not listed may be visited in between.
- The format of the routing header is



The extension header for routing.

- The first 4 bytes of the routing extension header contain four 1-byte integers.
- The **Next header** and **Header extension length** fields were described above.
- The **Routing type field** gives the format of the rest of the header.
 - Type 0 says that a reserved 32-bit word follows the first word, followed by some number of IPv6 addresses.
 - Other types may be invented in the future as needed.
- Finally, the **Segments left field** keeps track of how many of the addresses in the list have not yet been visited.
 - It is decremented every time one is visited.
 - When it hits 0, the packet is on its own with no more guidance about what route to follow.
 - Usually at this point it is so close to the destination that the best route is obvious.

- The **fragment header** deals with fragmentation similarly to the way IPv4 does.
- The header holds the **datagram identifier**, **fragment number**, and **a bit** telling whether more fragments will follow.
- In IPv6, unlike in IPv4, **only the source host** can fragment a packet. Routers along the way may not do this.
- if a router is confronted with a packet that is too big, it discards the packet and sends an **ICMP** packet back to the source.
- This information allows the **source host to fragment the packet** into smaller pieces using this header and try again.

- The **authentication header** provides a mechanism by which the receiver of a packet can be sure of who sent it.
- The encrypted security payload makes it possible to encrypt the contents of a packet so that only the intended recipient can read it.
- These headers **use cryptographic techniques** to accomplish their missions.

IPV4	IPV6
IPv4 has 32-bit address length	IPv6 has 128-bit address length
It Supports Manual and DHCP address configuration	It supports Auto and renumbering address configuration
In IPv4 end to end connection integrity is Unachievable	In IPv6 end to end connection integrity is Achievable
It can generate 4.29×10^9 address space	Address space of IPv6 is quite large it can produce 3.4×10^{38} address space
Security feature is dependent on application	IPSEC is inbuilt security feature in the IPv6 protocol
Address representation of IPv4 in decimal	Address Representation of IPv6 is in hexadecimal
Fragmentation performed by Sender and forwarding routers	In IPv6 fragmentation performed only by sender
In IPv4 Packet flow identification is not available	In IPv6 packetflow identification are Available and uses flow label field in the header
In IPv4 checksumfield is available	In IPv6 checksumfield is not available
It has broadcast Message Transmission Scheme	In IPv6 multicast and any cast message transmission scheme is available
In IPv4 Encryption and Authentication facility not provided	In IPv6 Encryption and Authentication are provided
IPv4 has header of 20-60 bytes.	IPv6 has header of 40 bytes fixed

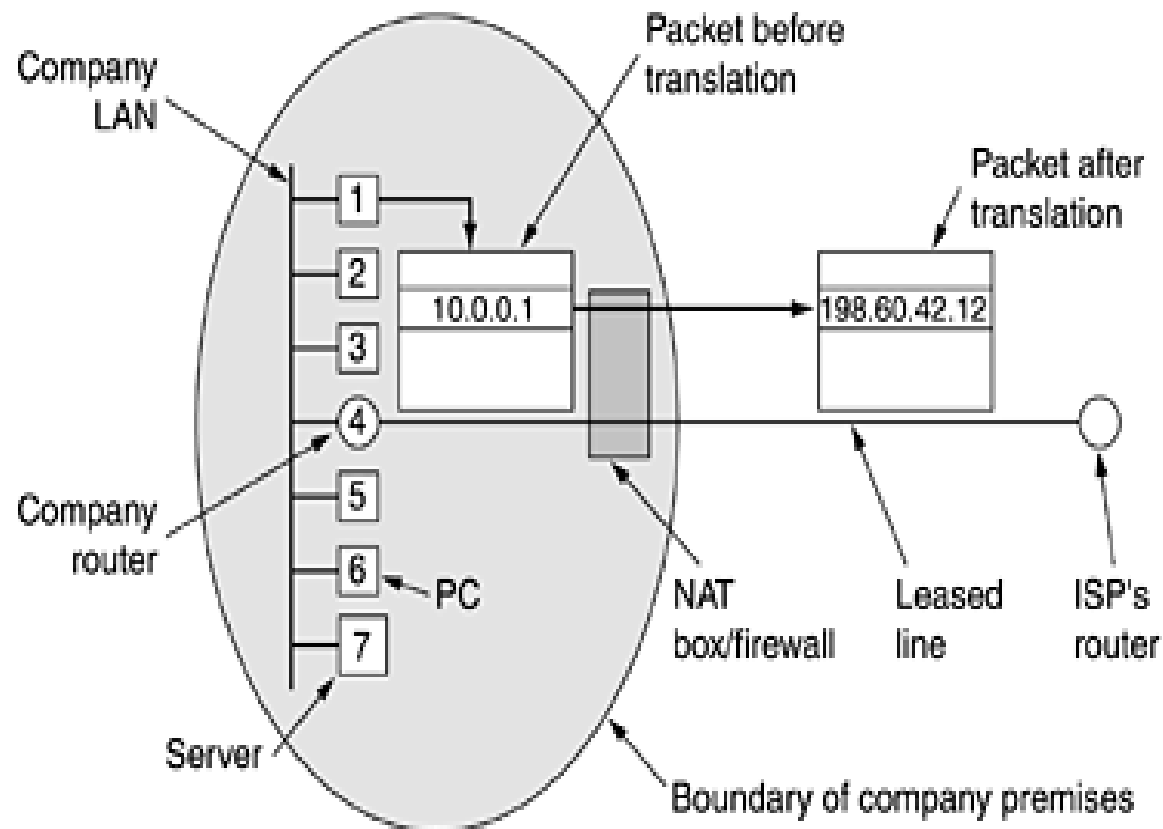
NAT-Network Address Translation

- The problem of running out of IP addresses is not a theoretical problem that might occur at some point in the distant future.
- The long-term solution is for the whole Internet to migrate to IPv6, which has 128-bit addresses.
- This transition is slowly occurring, but it will be years before the process is complete.
- This quick fix came in the form of NAT (Network Address Translation)
- The basic idea behind NAT is to **assign each company a single IP address** (or at most, a small number of them) for Internet traffic.
- Within the company, **every computer gets a unique IP address**, which is used for routing intramural traffic.

- To make this scheme possible, three ranges of IP addresses have been declared as private.
- Companies may use them internally as they wish. The only rule is that no packets containing these addresses may appear on the Internet itself. The three reserved ranges are:
 - 10.0.0.0 - 10.255.255.255/8 (16,777,216 hosts)
 - 172.16.0.0 - 172.31.255.255/12 (1,048,576 hosts)
 - 192.168.0.0 - 192.168.255.255/16 (65,536 hosts)
- The first range provides for 16,777,216 addresses (except for 0 and -1, as usual) and is the usual choice of most companies, even if they do not need so many addresses.

The operation of NAT

Figure 5-60. Placement and operation of a NAT box.



Here are some of the objections

- **First**, NAT violates the architectural model of IP, which states that every IP address uniquely identifies a single machine worldwide. The whole software structure of the Internet is built on this fact. With NAT, thousands of machines may (and do) use address 10.0.0.1.
- **Second**, NAT changes the Internet from a connectionless network to a kind of connection-oriented network. The problem is that the NAT box must maintain information (the mapping) for each connection passing through it. Having the network maintain connection state is a property of connection-oriented networks, not connectionless ones. If the NAT box crashes and its mapping table is lost, all its TCP connections are destroyed.

- **Third**, NAT violates the most fundamental **rule of protocol layering**: layer k may not make any assumptions about what layer $k + 1$ has put into the payload field. This basic principle is there to keep the layers independent.
- **Fourth**, processes on the Internet are not required to use TCP or UDP. If a user on machine A decides to use some new transport protocol to talk to a user on machine B (for example, for a multimedia application), introduction of a NAT box will cause the application to fail because the NAT box will not be able to locate the TCP Source port correctly.
- **Fifth**, some applications insert IP addresses in the body of the text. FTP, the standard File Transfer Protocol works this way and can fail in the presence of NAT unless special precautions are taken.
- **Sixth**, since the TCP Source port field is 16 bits, at most 65,536 machines can be mapped onto an IP address. Actually, the number is slightly less because the first 4096 ports are reserved for special uses. However, if multiple IP addresses are available, each one can handle up to 61,440 machines

Internet Control Protocols

- **IMCP—The Internet Control Message Protocol**
- **ARP—The Address Resolution Protocol**
- **DHCP—The Dynamic Host Configuration Protocol**

Label Switching and MPLS:

1. Internet Control Message Protocol

- The operation of the Internet is monitored closely by the routers. When something unexpected occurs during packet processing at a router, the event is reported to the sender by the **ICMP (Internet Control Message Protocol)**.
- **ICMP is also** used to test the Internet. About a dozen types of ICMP messages are defined.
- Each ICMP message type is carried encapsulated in an IP packet.

Internet Control Protocols (1)

Message type	Description
Destination unreachable	Packet could not be delivered
Time exceeded	Time to live field hit 0
Parameter problem	Invalid header field
Source quench	Choke packet
Redirect	Teach a router about geography
Echo and Echo reply	Check if a machine is alive
Timestamp request/reply	Same as Echo, but with timestamp
Router advertisement/solicitation	Find a nearby router

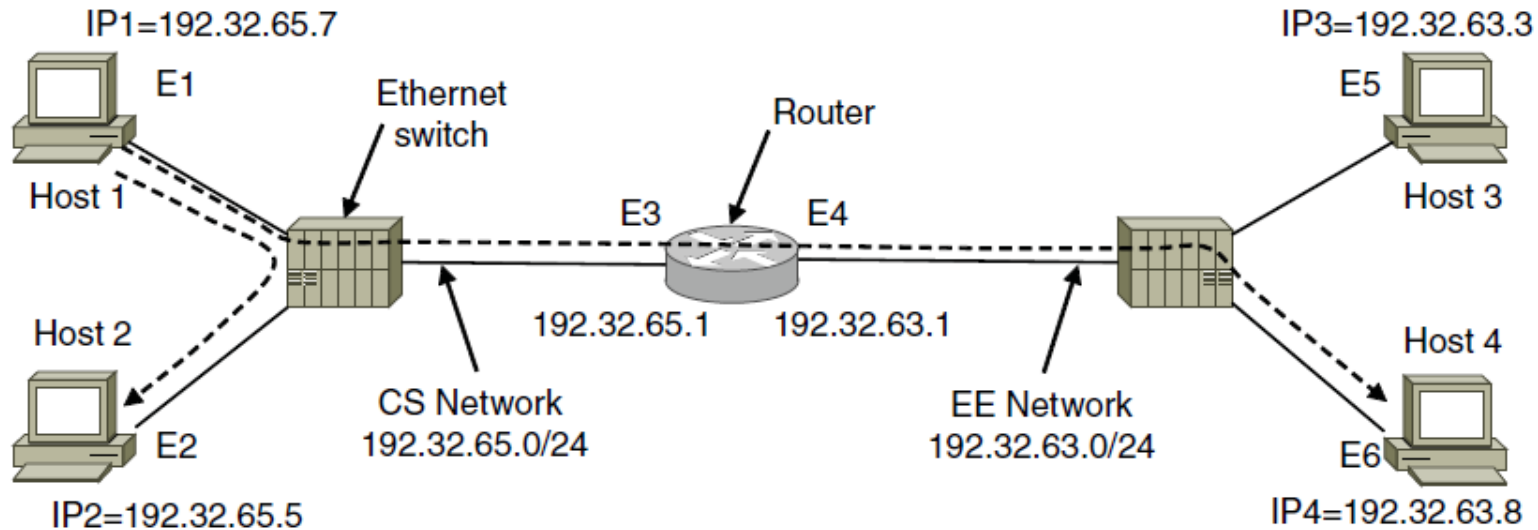
The principal ICMP (Internet Control Message Protocol) message types.

- The **DESTINATION UNREACHABLE** message is used when the **subnet or a router cannot locate the destination** or when a packet with the **DF bit** cannot be delivered because a "small-packet" network stands in the way.
- The **TIME EXCEEDED** message is sent when a packet is dropped because its **counter has reached zero**. This event is a **symptom** that packets are **looping**, that there is enormous **congestion**, or that the timer values are being set too **low**.
- The **PARAMETER PROBLEM** message indicates that an **illegal value has been detected in a header field**. This problem indicates a bug in the sending host's IP software or possibly in the software of a router transited.
- The **SOURCE QUENCH** message was formerly used to throttle hosts that were sending too many packets. When a host received this message, it was expected to slow down. It is rarely used any more because when congestion occurs.
- The **REDIRECT** message is used when a router **notices that a packet seems to be routed wrong**. It is used by the router to tell the sending host about the probable error.

2. ARP—The Address Resolution Protocol

- Data link layer NICs (Network Interface Cards) such as Ethernet cards do not understand Internet addresses.
- In the case of Ethernet, every NIC - equipped with a unique 48-bit Ethernet address.
- The NICs send and receive frames based on 48-bit Ethernet addresses. They know nothing at all about 32-bit IP addresses.
- The advantage of using ARP over configuration files is the simplicity.
- The system manager does not have to do much except assign each machine an IP address and decide about subnet masks. ARP does the rest.
- **Proxy ARP** is used in special cases in which a host wants to appear on a network even though it actually resides on another network.
- A common situation, for example, is a mobile computer that wants some other node to pick up packets for it when it is not on its home network.

Internet Control Protocols (2)



Frame	Source IP	Source Eth.	Destination IP	Destination Eth.
Host 1 to 2, on CS net	IP1	E1	IP2	E2
Host 1 to 4, on CS net	IP1	E1	IP4	E3
Host 1 to 4, on EE net	IP1	E4	IP4	E6

Two switched Ethernet LANs joined by a router

RARP

- **The Reverse Address Resolution Protocol (RARP)** is an obsolete computer networking protocol used by a client computer to request its Internet Protocol (IPv4) address from a computer network, when all it has available is its link layer or hardware address, such as a MAC address.
- ARP solves the **problem of finding out which Ethernet address corresponds to a given IP address**.
- Sometimes the **reverse problem** has to be solved: Given an Ethernet address, what is the corresponding IP address?
- In particular, this problem occurs **when a diskless workstation is booted**. Such a machine will normally get the binary image of its operating system from a remote file server.
- But how does it learn its IP address?

- The first solution devised was to use RARP (**Reverse Address Resolution Protocol**) (defined in RFC 903).
- This protocol allows a **newly-booted workstation to broadcast** its Ethernet address and say: My 48-bit Ethernet address is 14.04.05.18.01.25. **Does anyone out there know my IP address?**
- The **RARP server** sees this request, looks up the Ethernet address in its configuration files, and sends back the corresponding IP address.

Bootstrap

- A **disadvantage** of **RARP** is that it uses a destination address of all 1s (limited broadcasting) to reach the RARP server.
- However, such **broadcasts are not forwarded by routers**, so a RARP server is needed on each network.
- To get around this problem, an alternative **bootstrap** protocol called **BOOTP** was invented.

- Unlike RARP, BOOTP uses UDP messages, which are forwarded over routers. It also provides a diskless workstation with additional information, including the IP address of the file server holding the memory image, the IP address of the default router, and the subnet mask to use.
- A serious problem with BOOTP is that it requires manual configuration of tables mapping IP address to Ethernet address.
- When a new host is added to a LAN, it cannot use BOOTP until an administrator has assigned it an IP address and entered its (Ethernet address, IP address) into the BOOTP configuration tables by hand.

DHCP

(Dynamic Host Configuration Protocol)

- To eliminate this error-prone step, BOOTP was extended and given a new name: DHCP (Dynamic Host Configuration Protocol).
- DHCP allows both manual IP address assignment and automatic assignment.
- Like RARP and BOOTP, DHCP is based on the idea of a special server that assigns IP addresses to hosts asking for one.
- This server need not be on the same LAN as the requesting host. Since the DHCP server may not be reachable by broadcasting, a DHCP relay agent is needed on each LAN.

DHCP—The Dynamic Host Configuration Protocol

- ARP (as well as other Internet protocols) makes the assumption that hosts are configured with some basic information, such as their own IP addresses.
- How do hosts get this information? It is possible to manually configure each computer, but that is tedious and error-prone.
- There is a better way, and it is called **DHCP (Dynamic Host Configuration Protocol)**.
- With DHCP, every network must have a DHCP server that is responsible for configuration.
- when hosts do not have IP addresses, the server identifies a host using its Ethernet address (which is carried in the DHCP DISCOVER packet).
- An issue that arises with automatic assignment of IP addresses from a pool is for how long an IP address should be allocated. If a host leaves the network and does not return its IP address to the DHCP server, that address will be permanently lost.
- After a period of time, many addresses may be lost. To prevent that from happening, IP address assignment may be for a fixed period of time, a technique called **leasing**.

- DHCP is described in RFCs 2131 and 2132. It is widely used in the Internet to configure all sorts of parameters in addition to providing hosts with IP addresses.
- As well as in business and home networks, DHCP is used by ISPs to set the parameters of devices over the Internet access link, so that customers do not need to phone their ISPs to get this information. Common examples of the information that is configured include the network mask, the IP address of the default gateway, and the IP addresses of DNS and time servers.
- DHCP has largely replaced earlier protocols (called RARP and BOOTP) with more limited functionality.

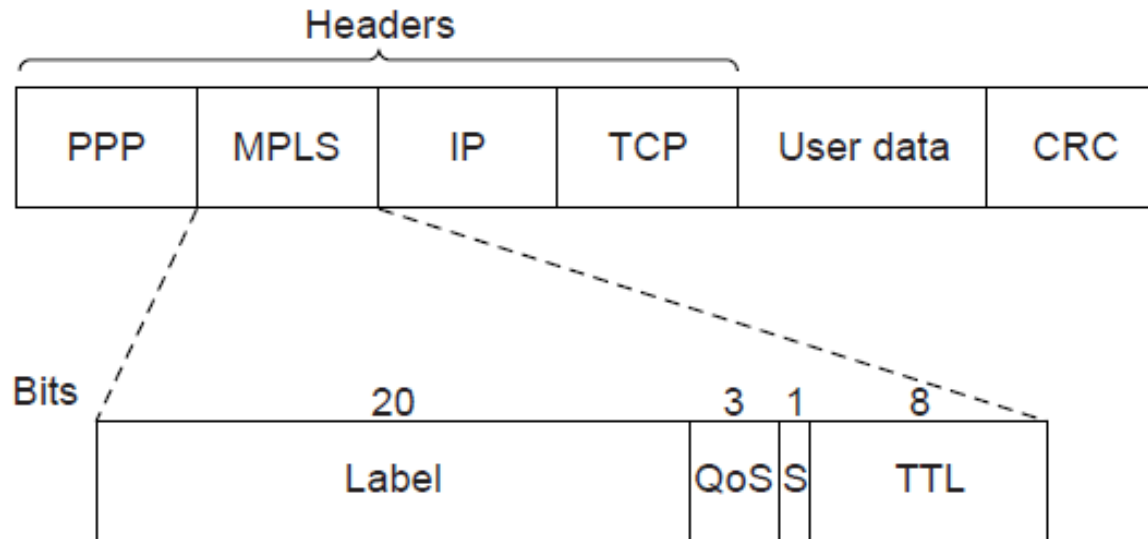
- To find its IP address, a newly-booted machine broadcasts a **DHCP DISCOVER** packet.
- The **DHCP relay** agent on its LAN intercepts all DHCP broadcasts.
- When it finds a DHCP DISCOVER packet, it sends the packet as a **unicast packet** to the **DHCP server**, possibly on a distant network.
- The only piece of information the relay agent needs is the IP address of the DHCP server.
- An issue that arises with **automatic assignment of IP addresses** from a pool is how long an IP address should be allocated.
- If a **host leaves the network and does not return its IP address** to the DHCP server, that address will be permanently lost.
- After a period of time, many addresses may be lost. To prevent that from happening, **IP address assignment may be for a fixed period of time**, a technique called **leasing**.
- Just before the lease expires, the host must ask the DHCP for a renewal. If it fails to make a request or the request is denied, the host may no longer use the IP address it was given earlier.

Label Switching and MPLS:

how packets are forwarded in the Internet.

- The network layer of the Internet, we have focused exclusively on packets as datagrams that are forwarded by IP routers.
- Another kind of technology that is starting to be widely used, especially by ISPs, in order to move Internet traffic across their networks. This technology is called **MPLS (MultiProtocol Label Switching)** and it is perilously close to circuit switching.
- MPLS adds a label in front of each packet, and forwarding is based on the label rather than on the destination address. Making the label an index into an internal table makes finding the correct output line just a matter of table lookup.
- Using this technique, **forwarding can be done very quickly**. This advantage was the original motivation behind MPLS, which began as proprietary technology known by various names including **tag switching**.

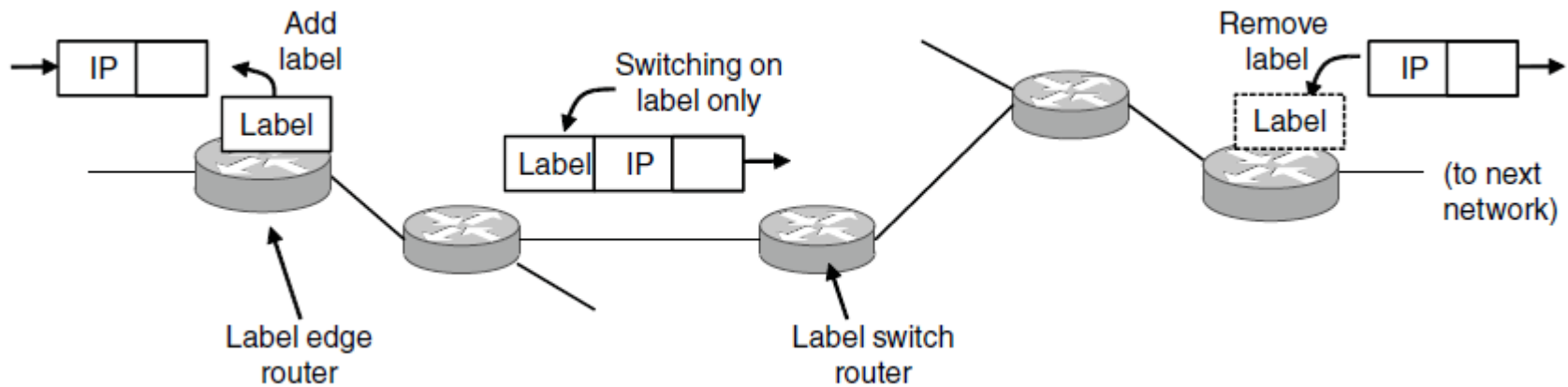
Label Switching and MPLS (1)



Transmitting a TCP segment using IP, MPLS, and PPP.

The generic MPLS header is 4 bytes long and has four fields. Most important is the *Label field*, which holds the index. The *QoS field* indicates the class of service. The *S field* relates to stacking multiple labels (which is discussed below). The *Ttl field* indicates how many more times the packet may be forwarded. It is decremented at each router, and if it hits 0, the packet is discarded. This feature prevents infinite looping in the case of routing instability.

Label Switching and MPLS (2)



Forwarding an IP packet through an MPLS network

- When an MPLS-enhanced packet arrives at a **LSR (Label Switched Router)**, the label is used as an index into a table to determine the outgoing line to use and also the new label to use. This label swapping is used in all virtual-circuit networks.
- Labels have only local significance and two different routers can feed unrelated packets with the same label into another router for transmission on the same outgoing line.
- **Forwarding** is the process of finding the best match for a destination address in a table to decide where to send packets.
- **Switching** uses a label taken from the packet as an index into a forwarding table. It is simpler and faster.

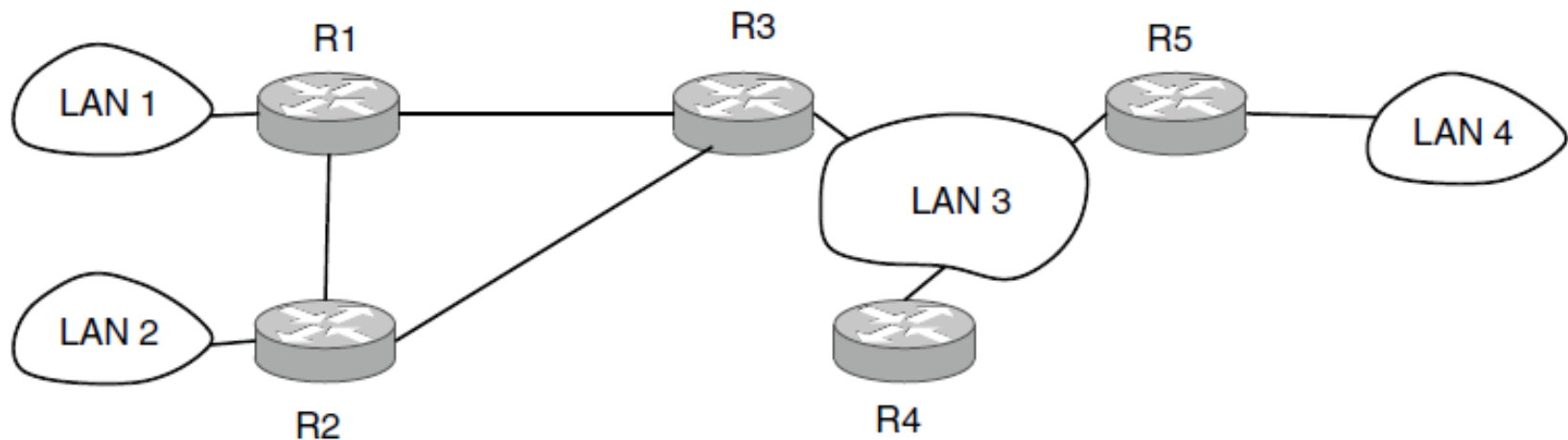
- When an IP packet reaches the edge of an MPLS network. The **LER (Label Edge Router) inspects** the destination IP address and other fields to see which MPLS path the packet should follow, and puts the right label on the front of the packet.
- Within the MPLS network, this label is used to forward the packet. At the other edge of the MPLS network, the label has served its purpose and is removed, revealing the IP packet again for the next network.
- The routers group multiple flows that end at a particular router or LAN and use a single label for them.
- The flows that are grouped together under a single label are said to belong to the same **FEC (Forwarding Equivalence Class)**.

OSPF—An Interior Gateway Routing Protocol:

Routing in the Internet

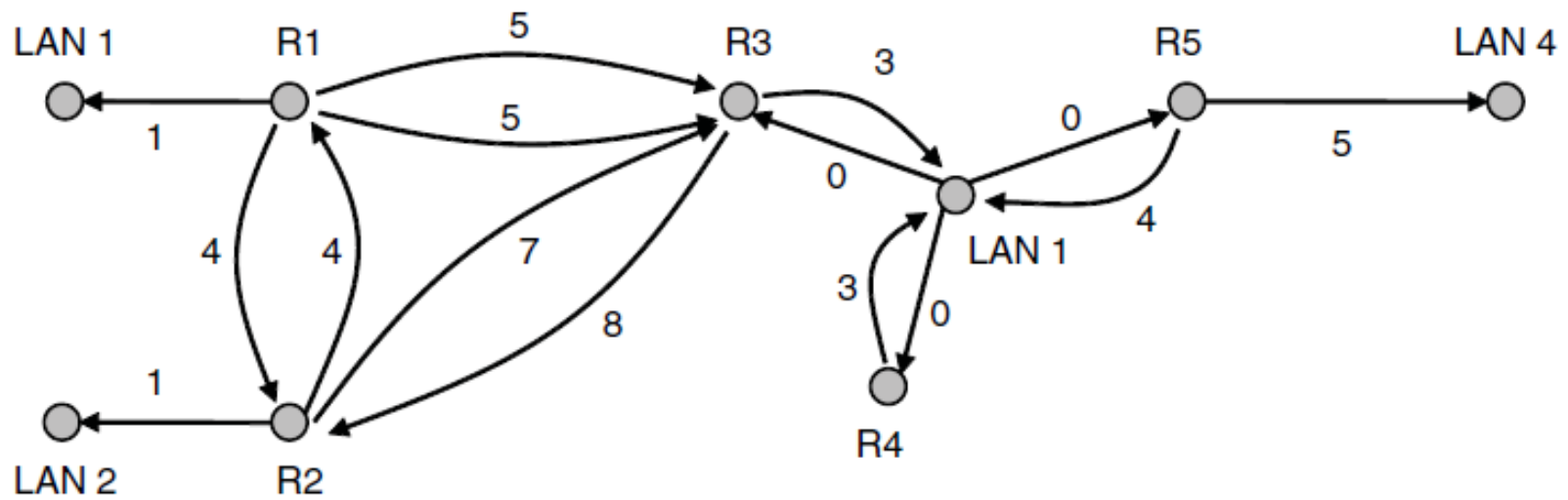
- The Internet is made up of a large number of independent networks or **ASes (Autonomous Systems)** that are operated by different organizations, usually a company, university, or ISP.
- Inside of its own network, an organization can use its own algorithm for internal routing, or **intradomain routing**, as it is more commonly known.
- An intradomain routing protocol is also called an **interior gateway protocol**.
- Early intradomain routing protocols used a distance vector design, based on the distributed Bellman-Ford algorithm inherited from the ARPANET.
- **RIP (Routing Information Protocol)** is the main example that is used to this day. It works well in small systems, but less well as networks get larger. It also **suffers from the count-to-infinity problem** and generally slow convergence.
- **OSPF (Open Shortest Path First)**- It drew on a protocol called **IS-IS (Intermediate-System to Intermediate-System)**, which became an ISO standard.
- **OSPF is more widely used in company networks, and IS-IS is more widely used in ISP networks.**
- OSPF supports both point-to-point links (e.g., SONET) and broadcast networks (e.g., most LANs).

OSPF—An Interior Gateway Routing Protocol (1)



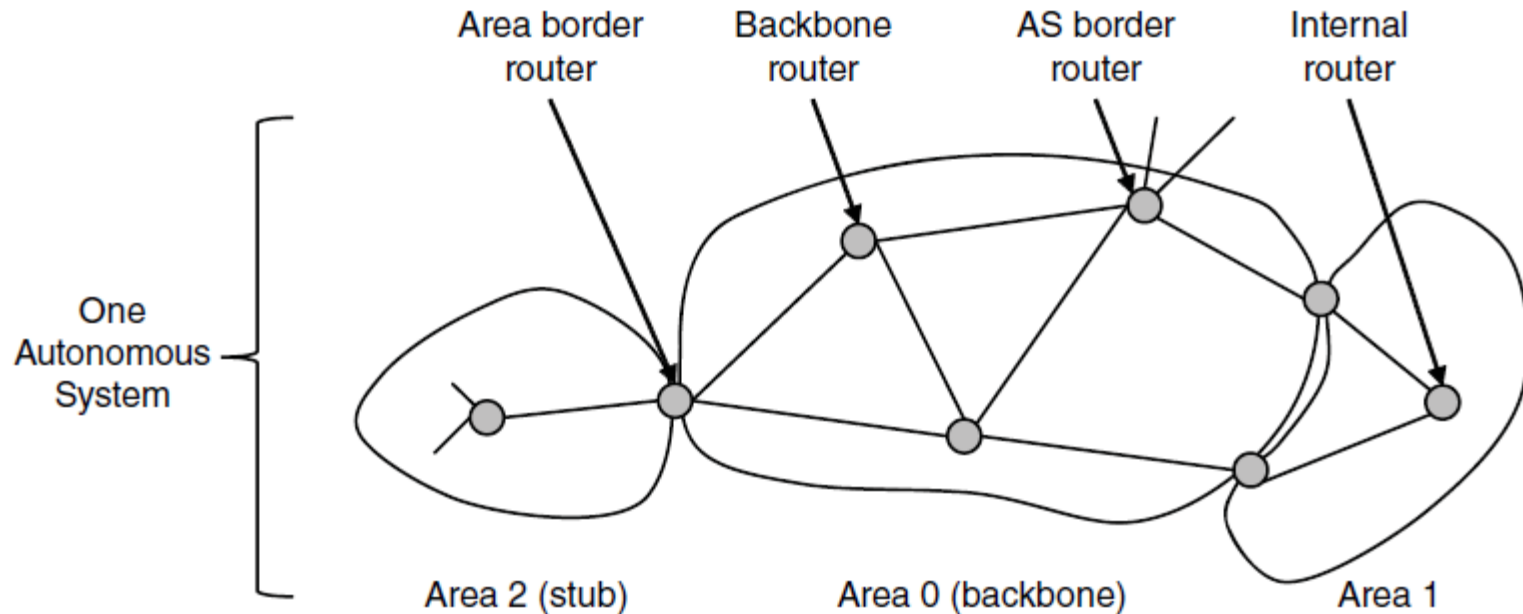
An autonomous system

OSPF—An Interior Gateway Routing Protocol (2)



A graph representation of the previous slide.

OSPF—An Interior Gateway Routing Protocol (3)



The relation between ASes, backbones, and areas in OSPF.

OSPF—An Interior Gateway Routing Protocol (4)

Message type	Description
Hello	Used to discover who the neighbors are
Link state update	Provides the sender's costs to its neighbors
Link state ack	Acknowledges link state update
Database description	Announces which updates the sender has
Link state request	Requests information from the partner

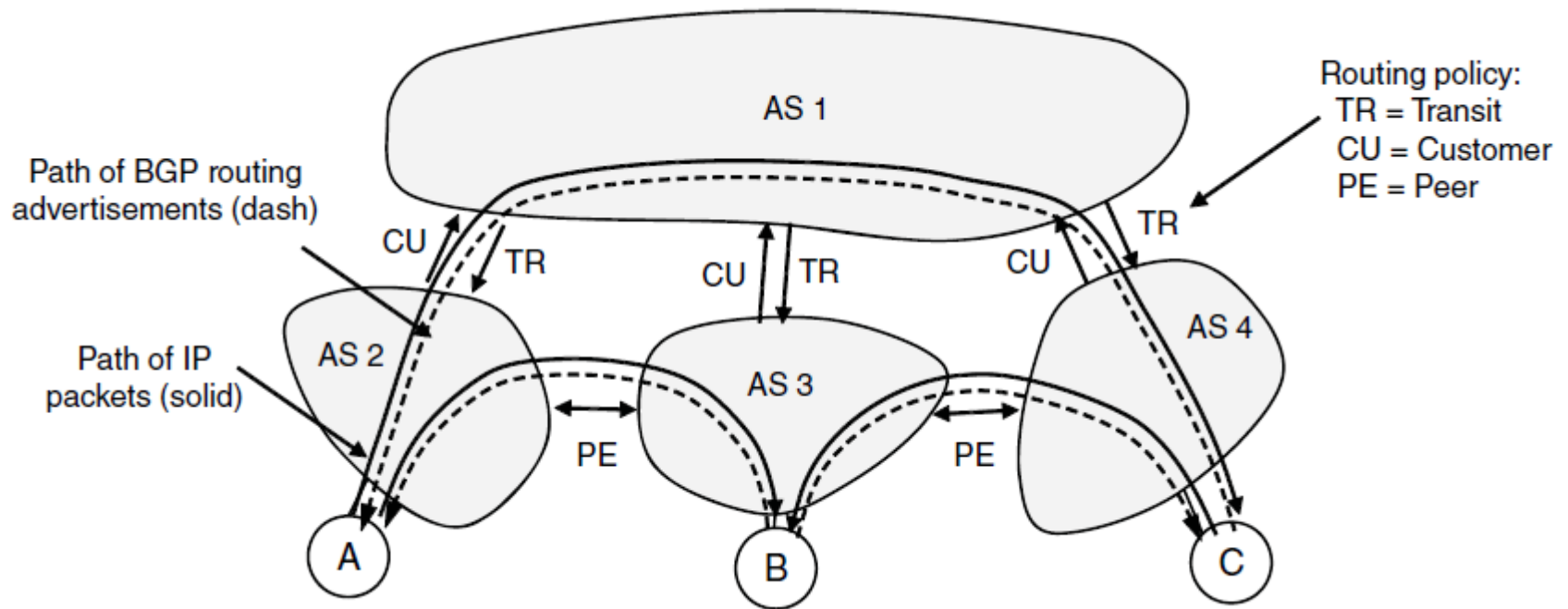
The five types of OSPF messages

BGP—The Exterior Gateway Routing Protocol (1)

Examples of routing constraints:

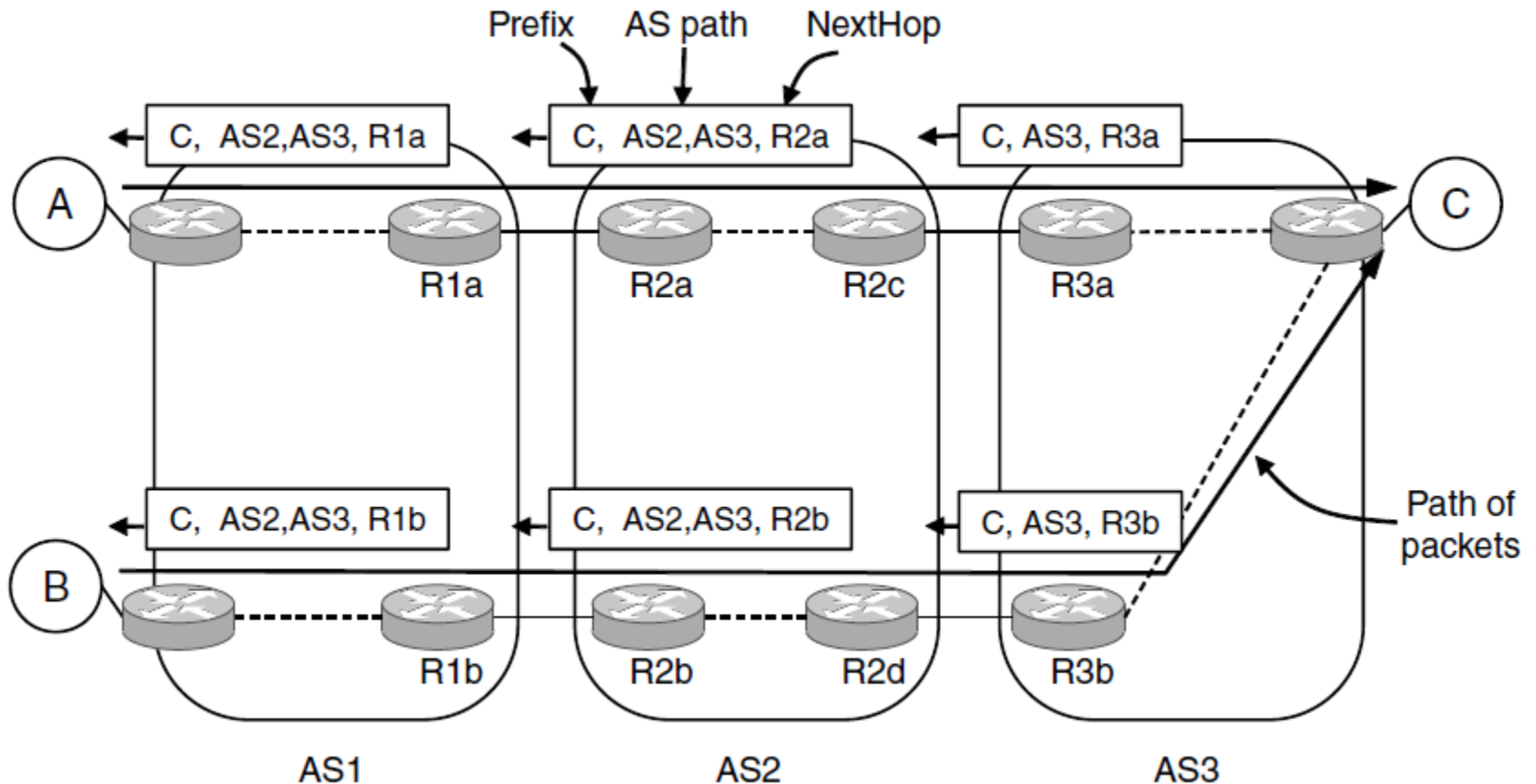
1. No commercial traffic for educat. network
2. Never put Iraq on route starting at Pentagon
3. Choose cheaper network
4. Choose better performing network
5. Don't go from Apple to Google to Apple

BGP—The Exterior Gateway Routing Protocol (2)



Routing policies between four Autonomous Systems

BGP—The Exterior Gateway Routing Protocol (3)



Propagation of BGP route advertisements

Mobile IP

Goals

1. Mobile host use home IP address anywhere.
2. No software changes to fixed hosts
3. No changes to router software, tables
4. Packets for mobile hosts – restrict detours
5. No overhead for mobile host at home.